

About Aerofit

Aerofit is a leading brand in the field of fitness equipment. Aerofit provides a product range including machines such as treadmills, exercise bikes, gym equipment, and fitness accessories to cater to the needs of all categories of people.

Business Problem

The market research team at AeroFit wants to identify the characteristics of the target audience for each type of treadmill offered by the company, to provide a better recommendation of the treadmills to the new customers. The team decides to investigate whether there are differences across the product with respect to customer characteristics.

Perform descriptive analytics to create a customer profile for each AeroFit treadmill product by developing appropriate tables and charts. For each AeroFit treadmill product, construct two-way contingency tables and compute all conditional and marginal probabilities along with their insights/impact on the business.

Dataset

The company collected the data on individuals who purchased a treadmill from the AeroFit stores during the prior three months. The dataset has the following features:

Dataset link: Aerofit_treadmill.csv

- **Product Purchased:** KP281, KP481, or KP781
- **Age:** In years
- **Gender:** Male/Female
- **Education:** In years **-MaritalStatus:** Single or partnered **-Usage:** The average number of times the customer plans to use the treadmill each week. **-Income:** Annual income (in \$) **-Fitness:** Self-rated fitness on a 1-to-5 scale, where 1 is the poor shape and 5 is the excellent shape. **-Miles:** The average number of miles the customer expects to walk/run each week

Product Portfolio:

- The KP281 is an entry-level treadmill that sells for \$1,500.
- The KP481 is for mid-level runners and sells for \$1,750.
- The KP781 treadmill has advanced features and sells for \$2,500.

Step 1: Import and Explore the Dataset

1.a.Import Libraries and Dataset

```
In [1]: import numpy as np
import pandas as pd
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv("/content/aerofit_treadmill.csv")
df.head(10)
```

Out[2]:

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47
5	KP281	20	Female	14	Partnered	3	3	32973	66
6	KP281	21	Female	14	Partnered	3	3	35247	75
7	KP281	21	Male	13	Single	3	3	32973	85
8	KP281	21	Male	15	Single	5	4	35247	141
9	KP281	21	Female	15	Partnered	2	3	37521	85

1.b.Checking the structure & characteristics of the dataset

Getting Shape of Data

```
In [3]: df.shape
```

Out[3]: (180, 9)

Getting Basic Dataset Info

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Product         180 non-null   object
1   Age             180 non-null   int64
2   Gender          180 non-null   object
3   Education        180 non-null   int64
4   MaritalStatus   180 non-null   object
5   Usage           180 non-null   int64
6   Fitness         180 non-null   int64
7   Income          180 non-null   int64
8   Miles           180 non-null   int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

Getting Datatypes Info

```
In [5]: df.dtypes
```

Out[5]:

Product	object
Age	int64
Gender	object
Education	int64

```
MaritalStatus    object
Usage            int64
Fitness          int64
Income           int64
Miles            int64
dtype: object
```

What are the Columns ?

```
In [6]: df.columns
```

```
Out[6]: Index(['Product', 'Age', 'Gender', 'Education', 'MaritalStatus', 'Usage',
              'Fitness', 'Income', 'Miles'],
              dtype='object')
```

The Column can be divided into 2 major types :

1. Categorical Columns:

- **Nominal Categorical:** "Product", "Gender", "MaritalStatus"
- **Ordinal Categorical:** "Fitness"

2. Numerical Columns : Age, Education, Usage, Income, Miles

Explanation:

Categorical Columns

- **Product:** Represents the product purchased (KP281, KP481, or KP781).
- **Gender:** Represents the gender of the user (Male or Female).
- **MaritalStatus:** Represents the marital status of the user (Single or Partnered).
- **Fitness:** Represents the self-rated fitness of the user on a scale from 1 to 5.

Numerical Columns

- **Age:** Represents the age of the user in years.
- **Education:** Represents the number of years of education.
- **Usage:** Represents the average number of times the customer plans to use the treadmill each week.
- **Income:** Represents the annual income of the user in dollars.
- **Miles:** Represents the average number of miles the customer expects to walk/run each week.

Converting Fitness to Object datatype :

```
In [7]: # Convert Fitness column to categorical (object)
        #df['Fitness'] = df['Fitness'].astype(str)
```

```
In [ ]: # Converting Int data type of fitness rating to object data type
        # df_cat = df
        # df_cat['Fitness_category'] = df.Fitness
        # df_cat.head()

        # df_cat["Fitness_category"].replace({1:"Poor Shape",
        #                                     2:"Bad Shape",
        #                                     3:"Average Shape",
        #                                     4:"Good Shape",
```

```
#                                     5: "Excellent Shape"}, inplace=True)
# df_cat.head()
```

Getting Statistical Data:

As the current dataset has numerical and categorical columns hence getting separate summary for both types

Summary statistics for numerical columns:

```
In [8]: df.describe()
```

Out[8]:

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

Summary statistics for non-numeric columns:

```
In [9]: df.describe(include=['object'])
```

Out[9]:

	Product	Gender	MaritalStatus
count	180	180	180
unique	3	2	2
top	KP281	Male	Partnered
freq	80	104	107

Observations:

Overall Count: 180

Demographics:

- The average age of AeroFit users is relatively young (28.79 years)
- 25% users are in 24 years range and 75% of the users are below 33 years
- Most users have a moderate to high level of education (mean of 15.57 years)

Product Preferences:

- The KP281 model is the most popular among users

Education:

- Mean Education is 15 years with maximum as 21 years and minimum as 12 years

Gender and Marital Status:

- Majority of users are male.
- More users are partnered than single.

Fitness Levels and Usage:

- The most common fitness level is 3.
- Users typically use the treadmill around 3.5 times per week and run approximately 103 miles per week.

Income:

- Users have a broad income range with an average around \$53,720, with minimum almost \$30K maximum of \$104K ,suggesting affordability across different income groups.

3.Checking the null values present in different columns

```
In [10]: df.isnull().sum()
```

```
Out[10]: Product      0
Age      0
Gender    0
Education 0
MaritalStatus 0
Usage     0
Fitness   0
Income    0
Miles     0
dtype: int64
```

Observation:

There are no nulls in the dataset

3.1.Non-Graphical Analysis: Value counts and unique attributes

```
In [ ]: # Getting "Counts of unique values" for a given column/feature :
```

```
print("\nCounts of unique values for 'Product':", df["Product"].value_counts())
print("\nCounts of unique values for 'Age':", df["Age"].value_counts())
print("\nCounts of unique values for 'Gender':", df["Gender"].value_counts())
print("\nCounts of unique values for 'Education':", df["Education"].value_counts())
print("\nCounts of unique values for 'MaritalStatus':", df["MaritalStatus"].value_counts())
print("\nCounts of unique values for 'Usage':", df["Usage"].value_counts())
print("\nCounts of unique values for 'Fitness':", df["Fitness"].value_counts())
print("\nCounts of unique values for 'Income':", df["Income"].value_counts())
print("\nCounts of unique values for 'Miles':", df["Miles"].value_counts())
```

```
In [ ]: # Getting "Unique values" for a given column/feature :
```

```
print("\nUnique values for 'Product':", df["Product"].unique())
print("\nUnique values for 'Age':", df["Age"].unique())
print("\nUnique values for 'Gender':", df["Gender"].unique())
print("\nUnique values for 'Education':", df["Education"].unique())
print("\nUnique values for 'MaritalStatus':", df["MaritalStatus"].unique())
```

```
print("\nUnique values for 'Usage':", df["Usage"].unique())
print("\nUnique values for 'Fitness':", df["Fitness"].unique())
print("\nUnique values for 'Income':", df["Income"].unique())
print("\nUnique values for 'Miles':", df["Miles"].unique())
```

```
In [ ]: # Getting "Number of unique values" for a given column/feature :

print("\nUnique values for 'Product':", df["Product"].nunique())
print("\nUnique values for 'Age':", df["Age"].nunique())
print("\nUnique values for 'Gender':", df["Gender"].nunique())
print("\nUnique values for 'Education':", df["Education"].nunique())
print("\nUnique values for 'MaritalStatus':", df["MaritalStatus"].nunique())
print("\nUnique values for 'Usage':", df["Usage"].nunique())
print("\nUnique values for 'Fitness':", df["Fitness"].nunique())
print("\nUnique values for 'Income':", df["Income"].nunique())
print("\nUnique values for 'Miles':", df["Miles"].nunique())
```

Summary:

1. **Product:** The most common product is KP281 (80), followed by KP481 (60) and KP781 (40).
2. **Age:** Ages range from 18 to 50, with the most frequent age being 25 (25 occurrences).
3. **Gender:** There are more males (104) than females (76).
4. **Education:** The most common years of education is 16 years (85 occurrences).
5. **Marital Status:** More respondents are partnered (107) than single (73).
6. **Usage:** The most frequent usage frequency is 3 times (69 occurrences).
7. **Fitness:** The most common fitness level is 3 (97 occurrences).
8. **Income:** Income varies widely, with the most common income being \$45,480 (14 occurrences).
9. **Miles:** The most common number of miles run is 85 miles (27 occurrences).

Step 2: Detect Outliers and Analyze Data Structure

2.a. Finding the outliers for every continuous variable in the dataset

Boxplot for numerical variables

```
num_cols = ['Age', 'Education', 'Usage', 'Income', 'Fitness', 'Miles']
```

```
In [14]: # List of numerical columns to create boxplots for
num_cols = ['Age', 'Education', 'Usage', 'Income', 'Fitness', 'Miles']

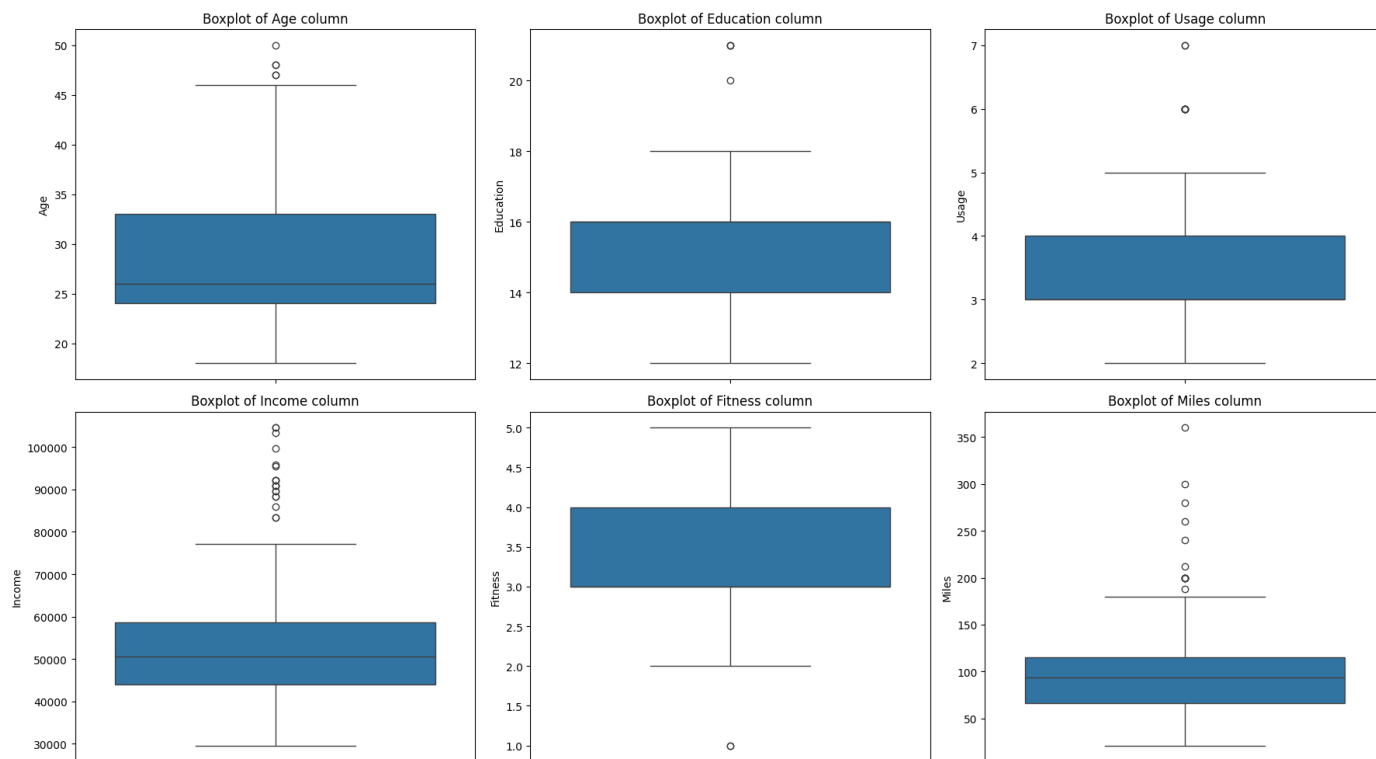
# Create subplots
fig, axes = plt.subplots(nrows=2, ncols=3, figsize=(18, 10))

# Flatten the axes array for easy iteration
axes = axes.flatten()

# Iterate over numerical columns and create a boxplot for each
for i, col in enumerate(num_cols):
    sns.boxplot(y=df[col], ax=axes[i])
    axes[i].set_title(f'Boxplot of {col} column')

# Adjust layout
```

```
plt.tight_layout()
plt.show()
```



Observations & Insights on all the Boxplots:

Boxplot	Observation	Insights
Age Column	- The median age is around 30 years. - The age distribution is fairly symmetric around the median. - There are a few outliers above the age of 45.	- The majority of customers are relatively young. - The presence of outliers indicates that a small number of older customers also purchase treadmills. - Marketing efforts could be tailored more towards the younger demographic but should also consider the needs of older customers.
Education Column	- The median education level is around 16 years. - The education distribution is slightly right-skewed. - There are a few outliers with education levels above 20 years.	- Most customers have a higher education level, suggesting they may be well-informed about fitness benefits. - The outliers indicate a few customers with very high education levels. - Educational content about fitness equipment could be effective in marketing.
Usage Column	- The median planned usage is around 3.5 i.e almost 4 times per week. - The distribution of usage is fairly symmetric with 25% and 50% coinciding - There are a few outliers with planned usage higher than 6 times per week.	- Most customers plan to use the treadmill moderately. - Some highly active users plan to use the treadmill extensively, indicating a potential market for more durable or advanced models.
Income Column	- The median income is around \$50,000 to \$60,000. - The income distribution is right-skewed. - There are numerous outliers with incomes above \$70,000.	- The majority of customers are in the middle-income range. - There is a significant portion of high-income customers, suggesting that premium products may have a viable market. - Pricing strategies should consider the income distribution to cater to both middle and high-income customers.
Fitness Column	- The median fitness rating is 3.	- Most customers rate their fitness at a

- The fitness distribution is symmetric.
- There is an outlier with a fitness rating of 1.

moderate level.

- The presence of a high fitness outlier suggests some customers consider themselves to be in excellent shape.
- Fitness level-based segmentation could help in targeting customers with specific product features.

Miles Column

- The median expected mileage is around 100 miles per week.
- The distribution of expected mileage is right-skewed.
- There are several outliers expecting to run more than 200 miles per week.

- Most customers have moderate mileage expectations.
- There is a notable segment of high-mileage runners, indicating a need for treadmills that can withstand extensive use.
- Offering products with higher durability and advanced features could appeal to these high-mileage customers.

2.b.Clipping Data Between the 5th and 95th Percentile

```
In [15]: # List of numerical columns to create boxplots for
num_cols = ['Age', 'Education', 'Usage', 'Income', 'Fitness', 'Miles']

for col in num_cols:
    lower_bound = df[col].quantile(0.05)
    upper_bound = df[col].quantile(0.95)
    df[col] = np.clip(df[col], lower_bound, upper_bound)
```

Step 3: Analyze Relationships and Patterns:

Check if features like marital status, Gender, and age have any effect on the product purchased

3.a.Univariate Analysis for Categorical Variables:

- **Histogram:** These can be used to visualize the distribution of continuous variables.
- **Countplot:** Generally used for categorical data, but could be used if you want to see counts of unique values .

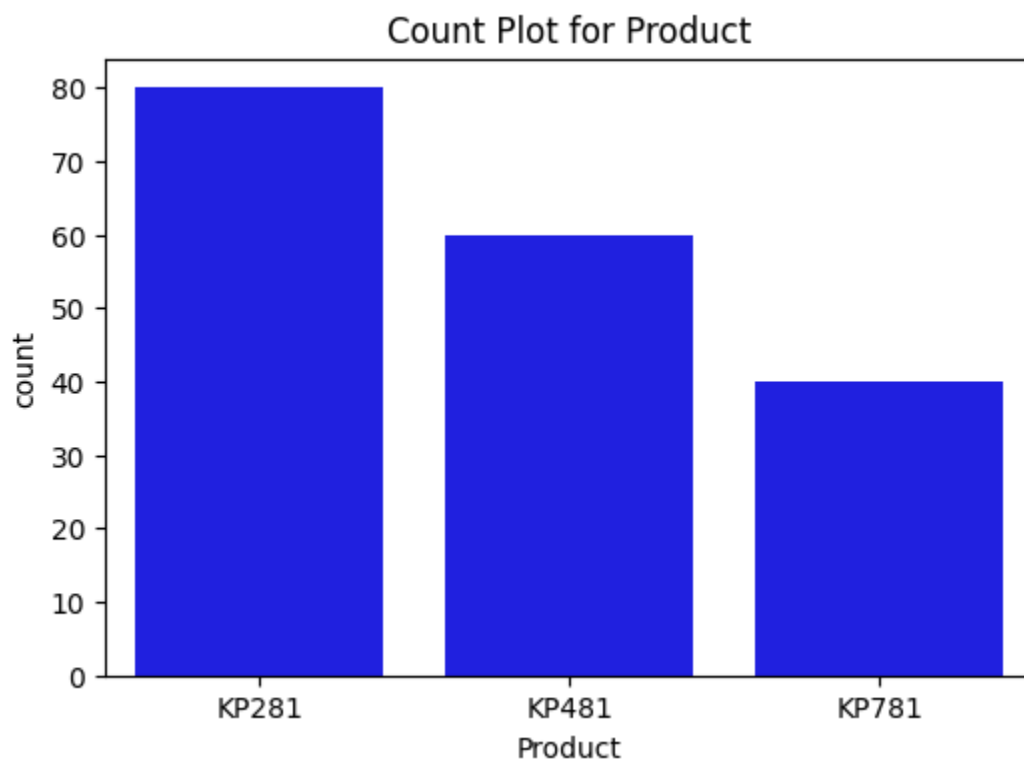
3.a.1.Feature/Variable "Product" :

```
In [16]: # Non Graphical Analysis
df["Product"].value_counts()
```

```
Out[16]: Product
KP281      80
KP481      60
KP781      40
Name: count, dtype: int64
```

```
In [17]: # Graphical Analysis

# Plot count plot for Product
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='Product', color="blue")
plt.title('Count Plot for Product')
plt.show()
```

Observations:

- **The KP281** is an entry-level treadmill that sells for **\$1,500** and seems to be the favorite as it is budget-friendly.
 - With its lower price point , it seems to attract a significant portion of buyers who might be looking for a more affordable option or are beginners in their fitness journey.
- **The KP481** is for mid-level runners and sells for **\$1,750**, making it the second choice.
 - This suggests that there is a considerable demand for treadmills with more features and capabilities than entry-level models but without the advanced functionalities of high-end models.
- **The KP781** treadmill has advanced features and sells for **\$2,500**, making it the costliest and least sold among all.
 - Despite its higher price point, it may appeal to a niche market segment seeking top-of-the-line features and willing to invest more in their fitness equipment.

Customer preferences show variety, with some opting for affordability and simplicity (**KP281**), while others prioritize advanced features and performance (**KP781**). This insight can be valuable for marketing strategies, product positioning, and inventory management.

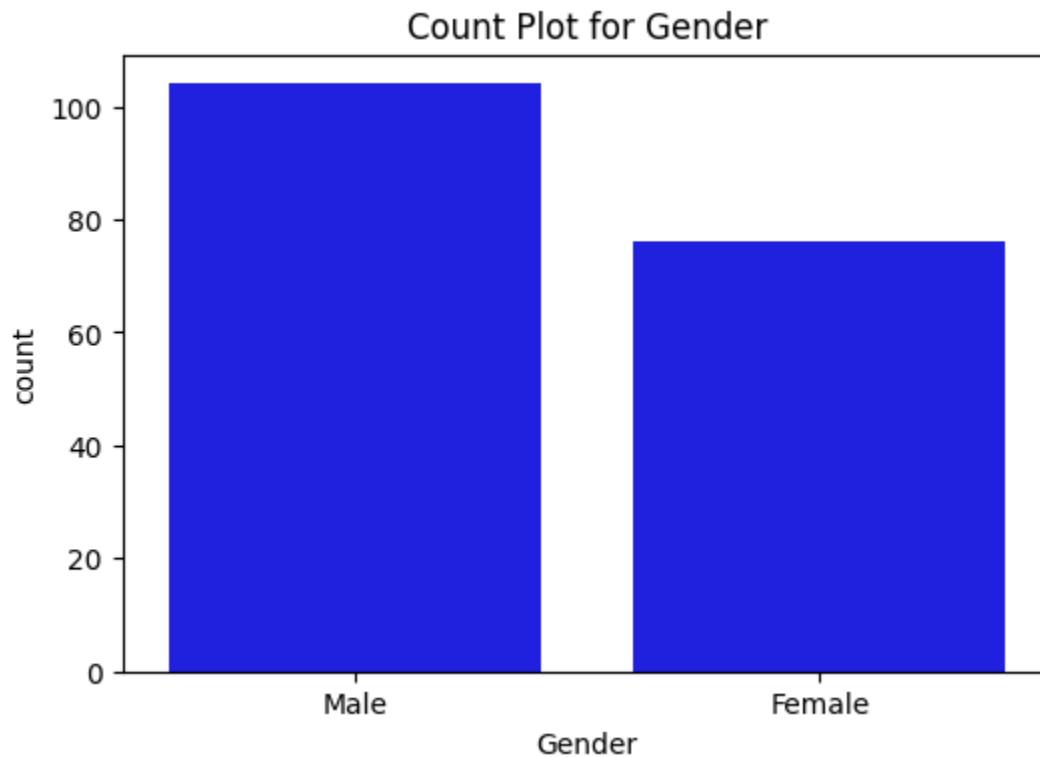
3.a.2.Feature/Variable "Gender" :

```
In [18]: # Non Graphical Analysis
df["Gender"].value_counts()
```

```
Out[18]: Gender
Male      104
Female     76
Name: count, dtype: int64
```

```
In [19]: # Graphical Analysis
```

```
# Plot count plot for Gender
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='Gender', color="blue")
plt.title('Count Plot for Gender')
plt.show()
```



Observation:

- The majority of customers are male, making up approximately 58% of the total customer base.
- Female customers constitute around 42% of the total customer base.

Business Implications:

- Marketing strategies should consider the larger male customer base, potentially focusing on male-preferred advertising channels and content.
- While the male customer base is larger, the significant female customer base (42%) should not be overlooked. Tailored marketing campaigns could be developed to better engage female customers.

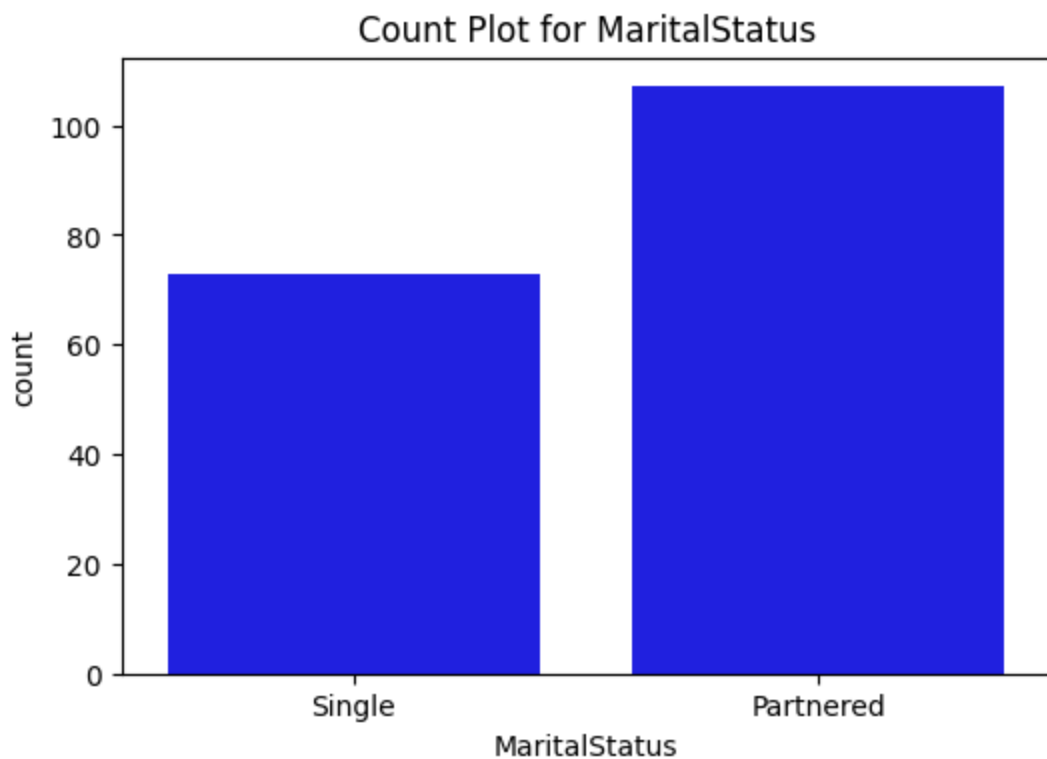
3.a.3.Feature/Variable "Marital Status" :

```
In [20]: # Non Graphical Analysis
df["MaritalStatus"].value_counts()
```

```
Out[20]: MaritalStatus
Partnered    107
Single        73
Name: count, dtype: int64
```

```
In [21]: # Graphical Analysis

# Plot count plot for MaritalStatus
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='MaritalStatus', color="blue")
plt.title('Count Plot for MaritalStatus')
plt.show()
```



Insights:

- The majority of customers are partnered, making up approximately 59.4% of the total customer base.
- Single customers constitute around 40.6% of the total customer base.

Business Implications:

- Marketing strategies should consider the larger partnered customer base, potentially focusing on family-oriented advertising channels and content.
- While the partnered customer base is larger, the significant single customer base (40.6%) should not be overlooked. Tailored marketing campaigns could be developed to better engage single customers, perhaps focusing on individual fitness goals and achievements.
- Promotional offers could include discounts for couples or family memberships, while also providing attractive offers for single customers to increase their engagement.

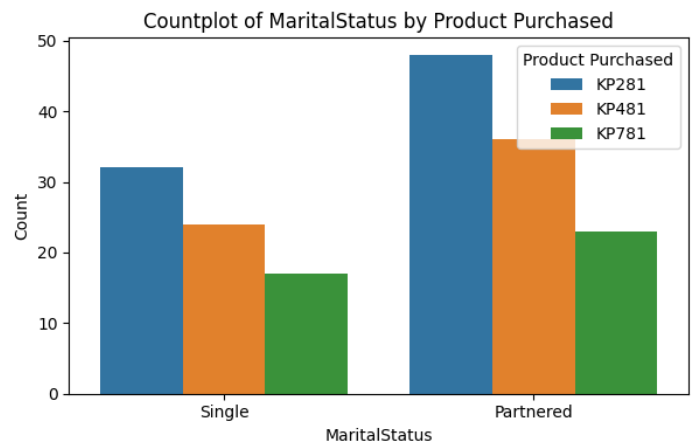
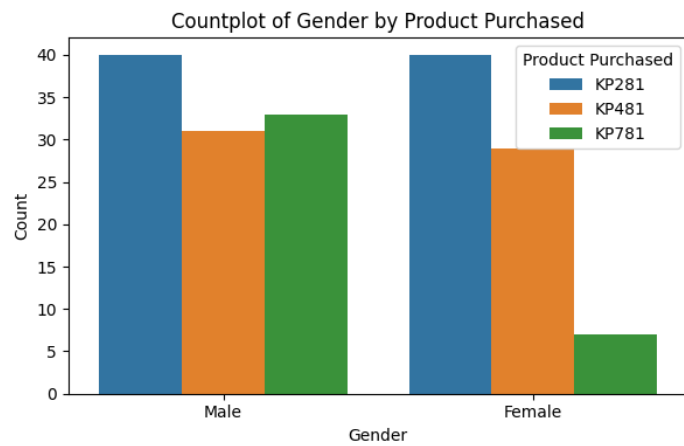
Check Relationships Between Categorical Variables and Product Purchased

```
In [22]: cat_cols = ['Gender', 'MaritalStatus']

fig, axes = plt.subplots(nrows=1, ncols=len(cat_cols), figsize=(12, 4))

for i, col in enumerate(cat_cols):
    sns.countplot(x=col, hue='Product', data=df, ax=axes[i])
    axes[i].set_title(f'Countplot of {col} by Product Purchased')
    axes[i].set_xlabel(col)
    axes[i].set_ylabel('Count')
    axes[i].legend(title='Product Purchased', loc='upper right')

plt.tight_layout()
plt.show()
```



Observations for Gender Preferences for Particular Model:

- The KP281 treadmill is slightly more popular among males than females.
- The KP481 treadmill has more male buyers compared to female buyers, but the difference is not as pronounced as with KP281
- The KP781 treadmill is significantly more popular among males compared to females

Observations for Marital Status for Particular Model:

- The KP281 treadmill is more popular among partnered individuals compared to single individuals
- The KP481 treadmill is also more popular among partnered individuals, but the difference is less pronounced compared to KP281
- The KP781 treadmill is more popular among partnered individuals, similar to the other models, but it has the lowest overall sales among single individuals

Summary

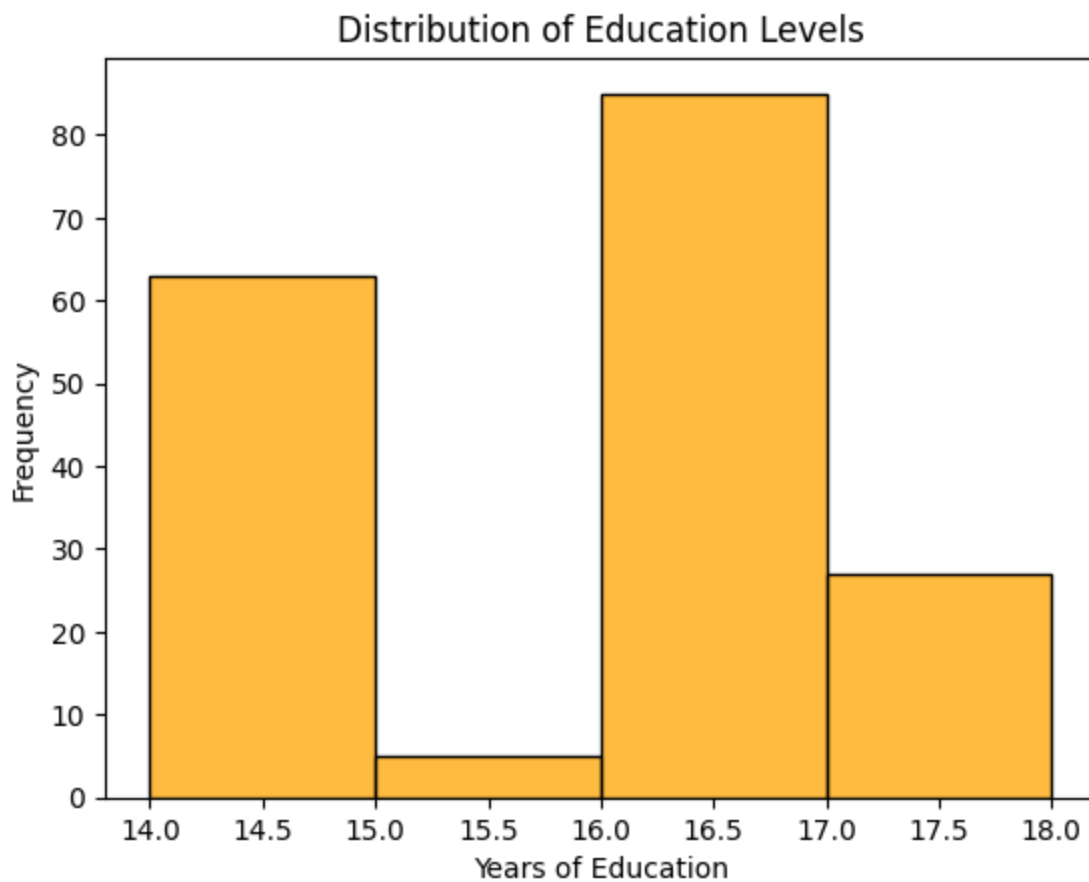
- **Gender Insight:** The KP281 and KP481 treadmills are relatively popular among both genders, but the KP781 treadmill shows a strong preference among males.
- **Marital Status Insight:** All three treadmill models are more popular among partnered individuals compared to single individuals, with KP281 showing the strongest preference among partnered customers

Histograms/Kde for Numerical Variables:

- **Histplot for Discrete Variables:** Education, Fitness
- **KDEplot for Continuous Variables:** Age, Income, Usage, Miles

3.a.4.Feature/Variable "Education" :

```
In [23]: sns.histplot(data=df, x='Education', kde=False, bins=4, color='orange')
plt.title('Distribution of Education Levels')
plt.xlabel('Years of Education')
plt.ylabel('Frequency')
plt.show()
```



Observations:

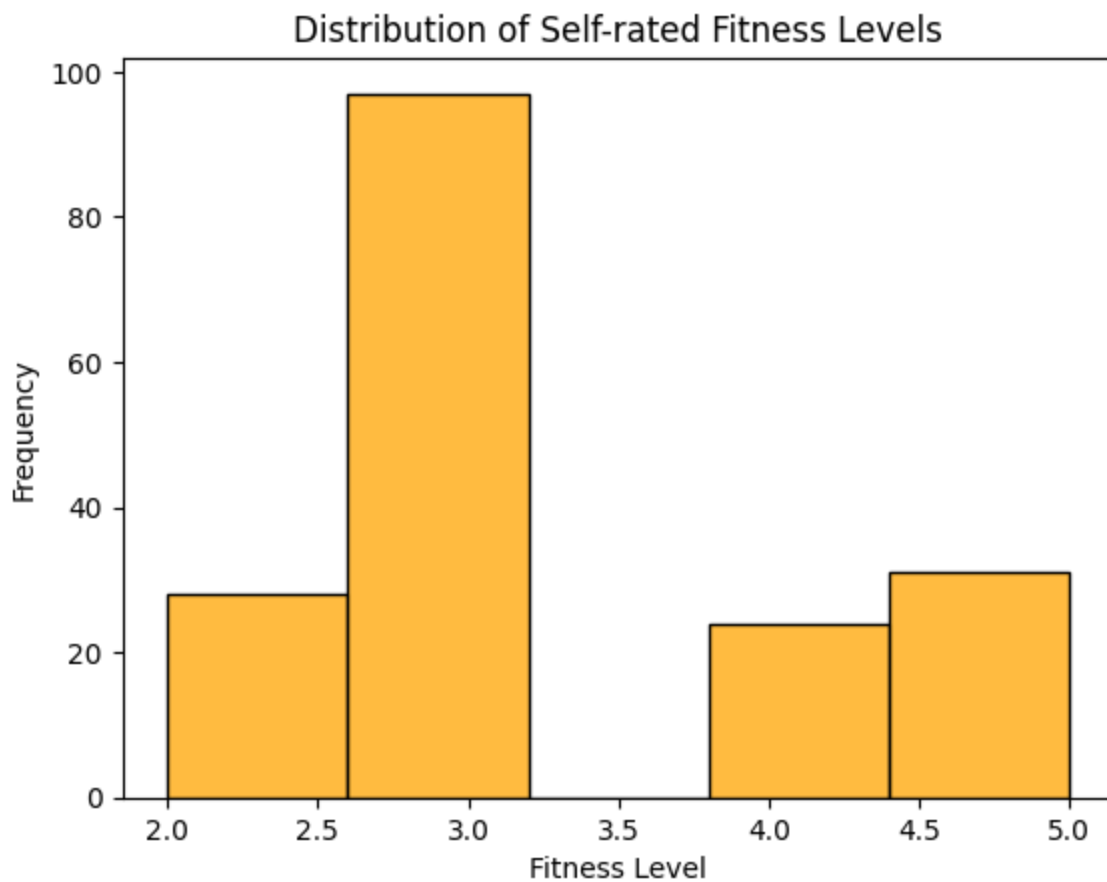
- The years of education in the dataset range from 14 to 18 years.
- The histogram shows a bimodal distribution with peaks around 14 and 16 years of education. This suggests that there are two common education levels among the customers: those with around 14 years (likely high school graduates) and those with around 16 years (likely college graduates).

Business Insights:

- **Target Audience:** Most customers have either high school or college-level education. Tailor marketing strategies to these groups.
- **Educational Programs:** Consider introducing fitness-related educational or training programs to engage these customers.
- **Product Customization:** Adjust product information and promotional materials based on education level, offering more detailed information to those with higher education.

3.a.5.Feature/Variable "Fitness" :

```
In [24]: # Fitness
sns.histplot(data=df, x='Fitness', kde=False, bins=5, color='orange')
plt.title('Distribution of Self-rated Fitness Levels')
plt.xlabel('Fitness Level')
plt.ylabel('Frequency')
plt.show()
```



Observations:

- The 'Fitness' attribute ranges from 1 to 5. Still nobody has rated themselves as 1.
- In the histogram, we see fitness levels from 2.0 to 5.0.
- The most common fitness level is 3.0, with a frequency of nearly 100, indicating many customers rate their fitness as average.
- Fitness level 2.0 has a smaller frequency, followed by levels 4.0 and 5.0, showing fewer customers rate their fitness as below or above average.
- There is a significant drop in frequency for fitness levels above 3.0, indicating fewer customers consider themselves highly fit.

Business Insights:

Product Development:

- Focus on treadmills that appeal to customers with average fitness (level 3.0).
- Introduce features or programs to help customers improve their fitness levels.

Marketing Strategies:

- Highlight how treadmills can improve the fitness of average customers.
- Use testimonials and success stories to promote the brand.

Customer Segmentation:

- Segment customers by fitness levels and design targeted campaigns.
- Offer special deals for those looking to improve from fitness levels 2.0 to 3.0 or 3.0 to 4.0.
- Target high-value customers (fitness levels 4.0 and 5.0) with premium products and services.

Fitness Programs:

- Develop programs or challenges that encourage customers to improve their fitness levels, fostering continuous engagement with the AeroFit brand.

3.a.6.Kde Plot for Features/Variables 'Age', 'Income', 'Usage', 'Miles' :

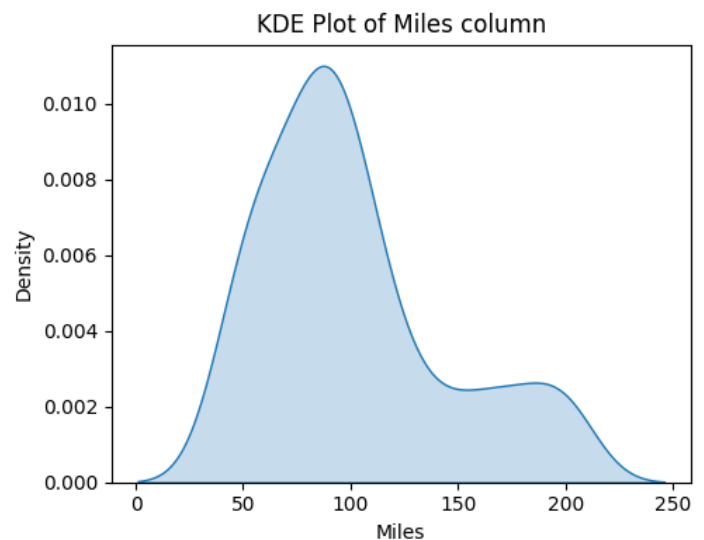
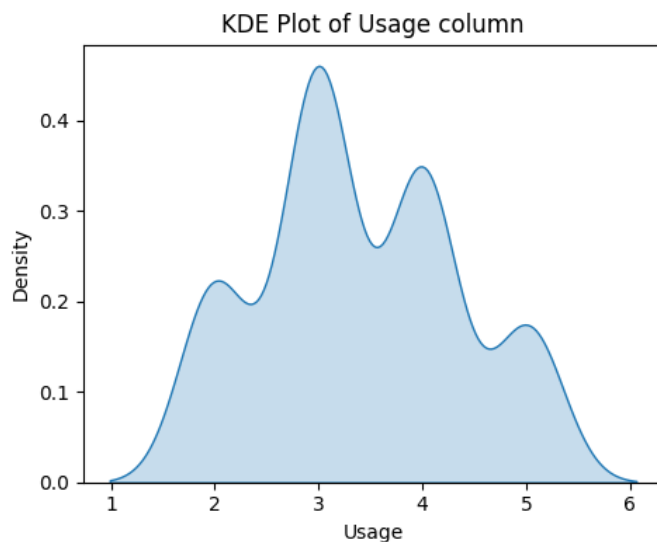
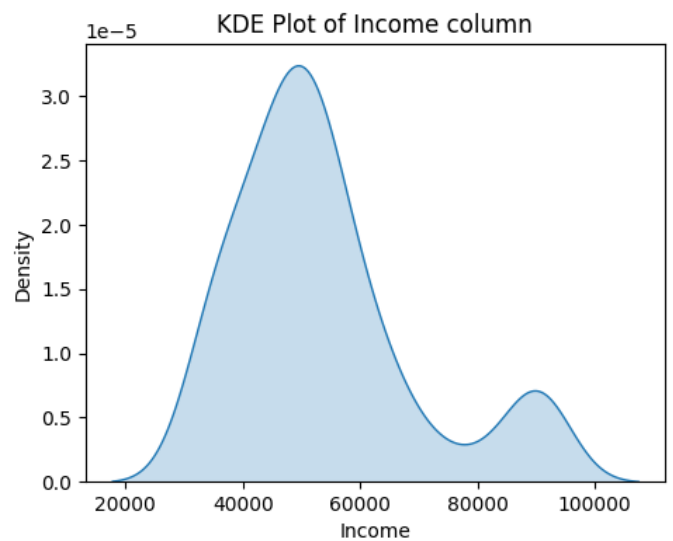
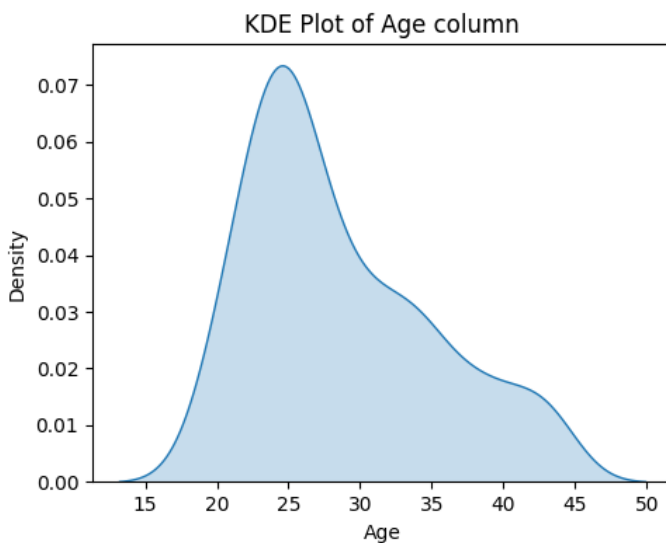
```
In [25]: # List of numerical columns for KDE plots
num_cols = ['Age', 'Income', 'Usage', 'Miles']

# Create subplots
fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(10, 8))

# Flatten the axes array for easy iteration
axes = axes.flatten()

# Iterate over numerical columns and create a KDE plot for each
for i, col in enumerate(num_cols):
    sns.kdeplot(data=df[col], ax=axes[i], fill=True)
    axes[i].set_title(f'KDE Plot of {col} column')

# Adjust layout
plt.tight_layout()
plt.show()
```



Category	Observations	Business Insights
Age	- Range: 15 to 50 years	- Target marketing efforts on younger

	- Peak around 25 years	customers, particularly those around 25 years old. - Design and develop products that cater to the preferences and needs of younger individuals.
Income	- Range: 20,000 to 100,000 - Peak around 50,000 – <i>Secondary peak around 90,000</i>	- Offer a variety of treadmills at different price points to cater to both mid-income (50,000) and high-income (90,000) customers. - Develop different models with varying features and pricing to appeal to these distinct income segments.
Usage	- Range: 1 to 6 times per week - Multiple peaks with highest around 3 times and significant at 5 times	- Segment customers based on their usage frequency (casual users: 1-3 times/week; frequent users: 4-6 times/week). - Offer different programs or features tailored for casual users and frequent users to enhance customer satisfaction and engagement.
Miles	- Range: 0 to 250 miles per week - Peak around 50 miles - Long tail for higher mileages	- Tailor marketing and product features for high-mileage users. - Emphasize durability, performance, and advanced tracking capabilities.

Fitness Programs

- Develop programs catering to different usage levels.
- Create mileage-based challenges and reward programs.
- Encourage customer engagement and loyalty through targeted fitness programs.
- Leverage data to personalize and enhance user experience.

For Fine details to be captured :Using Histplot with Kde and more number of bins- for all the continuous variables:

```
In [64]: # List of continuous columns
continuous_cols = ['Age', 'Education', 'Usage', 'Income', 'Fitness', 'Miles']

# Create subplots with 3 rows and 2 columns
fig, axes = plt.subplots(3, 2, figsize=(15, 15))

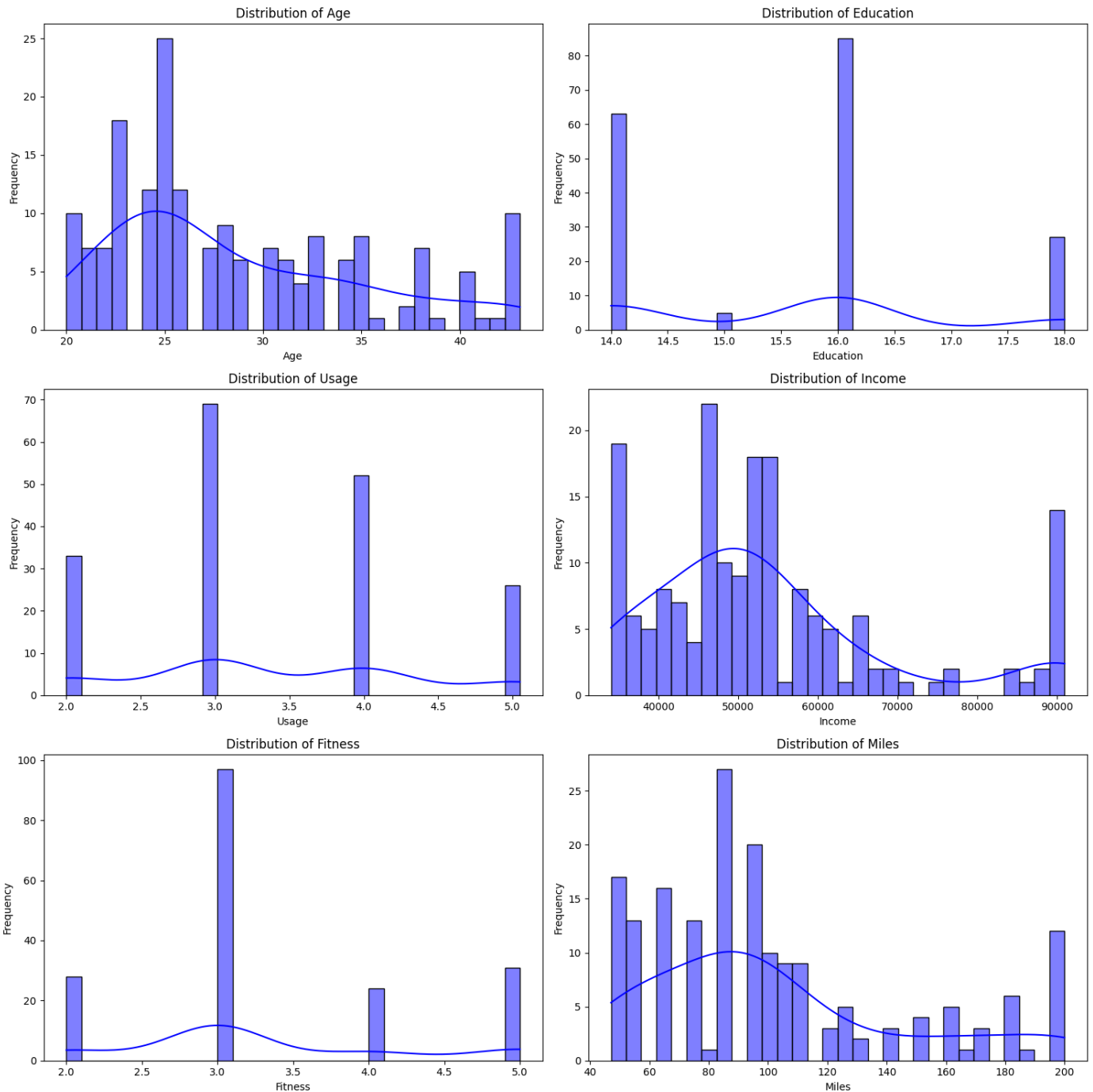
# Flatten the axes array for easy iteration
axes = axes.flatten()

# Plot distribution for each continuous variable
for i, col in enumerate(continuous_cols):
    sns.histplot(df[col], kde=True, bins=30, color='blue', ax=axes[i])
    axes[i].set_title(f'Distribution of {col}')
    axes[i].set_xlabel(col)
    axes[i].set_ylabel('Frequency')

# Hide any unused subplots
for j in range(i + 1, len(axes)):
    fig.delaxes(axes[j])

# Adjust layout
plt.tight_layout()

plt.show()
```

3.b.Bivariate Analysis Using Boxplots:

Selecting only the numerical columns for correlation analysis

```
numerical_columns = df.select_dtypes(include=['float64', 'int64']).columns
```

```
In [27]: # List of numerical columns
num_cols = ['Age', 'Education', 'Usage', 'Income', 'Fitness', 'Miles']

# List of categorical columns
cat_cols = ['Product', 'Gender', 'MaritalStatus']

# Set up the subplots
fig, axes = plt.subplots(nrows=2, ncols=3, figsize=(15, 10))
```

```

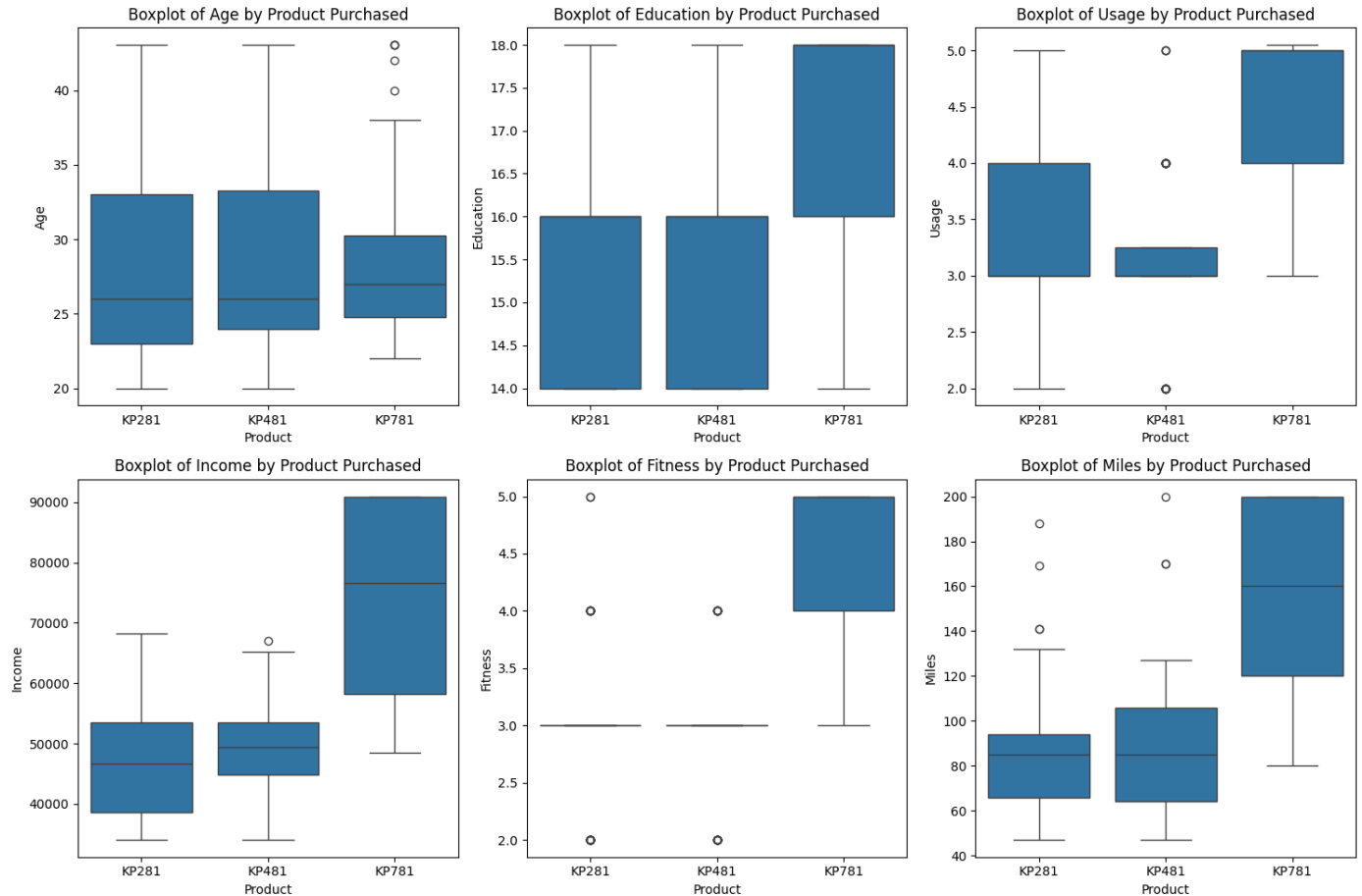
# Flatten the axes for easy iteration
axes = axes.flatten()

# Iterate through numerical columns
for i, col in enumerate(num_cols):
    sns.boxplot(x='Product', y=col, data=df, ax=axes[i])
    axes[i].set_title(f'Boxplot of {col} by Product Purchased')

# Remove any unused axes
for j in range(len(num_cols), len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()

```



Product	Age Observations	Education Observations	Usage Observations	Business Insights
KP281	- Median age: ~28 - Age range: 22-42 - Outliers: Above 45	- Median education: ~15 years - Education range: 14-18 years	- Median usage: ~4 times/week - Usage range: 2-5 times/week - Outliers: Below 2	- Appeals to a broad age group - General-purpose product - Consistent usage
KP481	- Median age: ~30 - Age range: 20-45 - Outliers: Above 45	- Median education: ~15 years - Education range: 14-18 years	- Median usage: ~2.5 times/week - Usage range: Narrow, consistent but low usage	- Broad age range appeal - Requires improvements to increase usage - Suitable for a general demographic
KP781	- Median age: ~27 - Age range: 22-35	- Median education: ~16 years - Education range: 14-18 years	- Median usage: ~4.5 times/week - Usage range: 3-5 times/week	- Appeals to younger, more educated consumers - Highest usage rate - Effective and satisfying product

Overall Business Insights

• Product Differentiation:

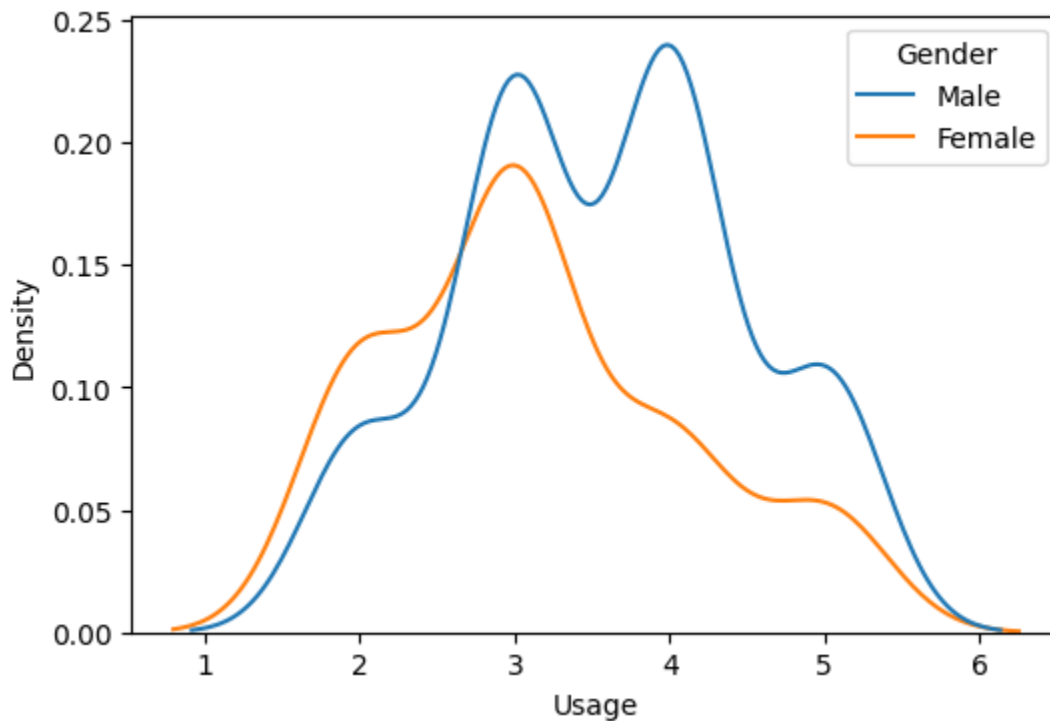
- KP781 is distinguished by younger users, higher education levels, and higher usage rates. It might be marketed as a high-end or specialized product.
- KP481 is suitable for a broad age range but has lower usage. Addressing potential improvements or better matching it with user needs could enhance its market position.
- KP281 attracts a diverse age group with moderate education levels and consistent usage, making it a solid general-purpose product.

Category	Observations	Business Insights
Income by Product Purchased		
KP281	- Most customers have an income between \$42,000 and \$57,000.	- Target middle-income customers with promotions and financing options.
	- Average income around \$50,000.	- Differentiate features from KP481 to target specific sub-groups.
KP481	- Average income is slightly lower at about \$48,000.	- Target similar income group as KP281 with a focus on different features or usage scenarios.
	- Most incomes range from \$41,000 to \$55,000.	
KP781	- Customers generally have higher incomes, averaging around \$80,000.	- Target wealthier individuals with marketing highlighting advanced features and benefits.
	- Most incomes range from \$60,000 to \$90,000.	
Fitness by Product Purchased		
KP281 and KP481	- Customers generally rate their fitness level at 3 out of 5, which is average.	- Promote as user-friendly and suitable for beginners or intermediate users.
	- Few customers rate their fitness much higher or lower.	- Offering fitness programs and support for beginners can enhance appeal.
KP781	- Customers tend to rate their fitness higher, with an average rating of 4 out of 5.	- Appeals to fitness-conscious individuals. Market with a focus on high performance and advanced features.
	- There is a mix of different fitness levels among these customers.	
Miles by Product Purchased		
KP281	- Customers expect to run or walk about 85 miles per week on average.	- Highlight reliability and user-friendly features for moderate usage.
	- Some plan to use it much more.	
KP481	- Customers expect to use it a bit more, around 90 miles per week on average.	- Similar insights to KP281, with a focus on consistent moderate usage.
KP781	- Customers plan to use it the most, around 140 miles per week on average.	- Target serious runners or walkers. Market durability and performance for heavy use.
	- Likely attracts serious runners or walkers.	- Customizing features to accommodate different usage levels can help meet the diverse needs of customers.

Relation between Product purchased Customers Usage per week and their Gender:

--

```
In [66]: plt.figure(figsize=(6,4))
sns.kdeplot(data=df,x='Usage',hue='Gender')
plt.show()
```



Observation: Most women plan on using the treadmill for lesser number of times per week as compared to men .

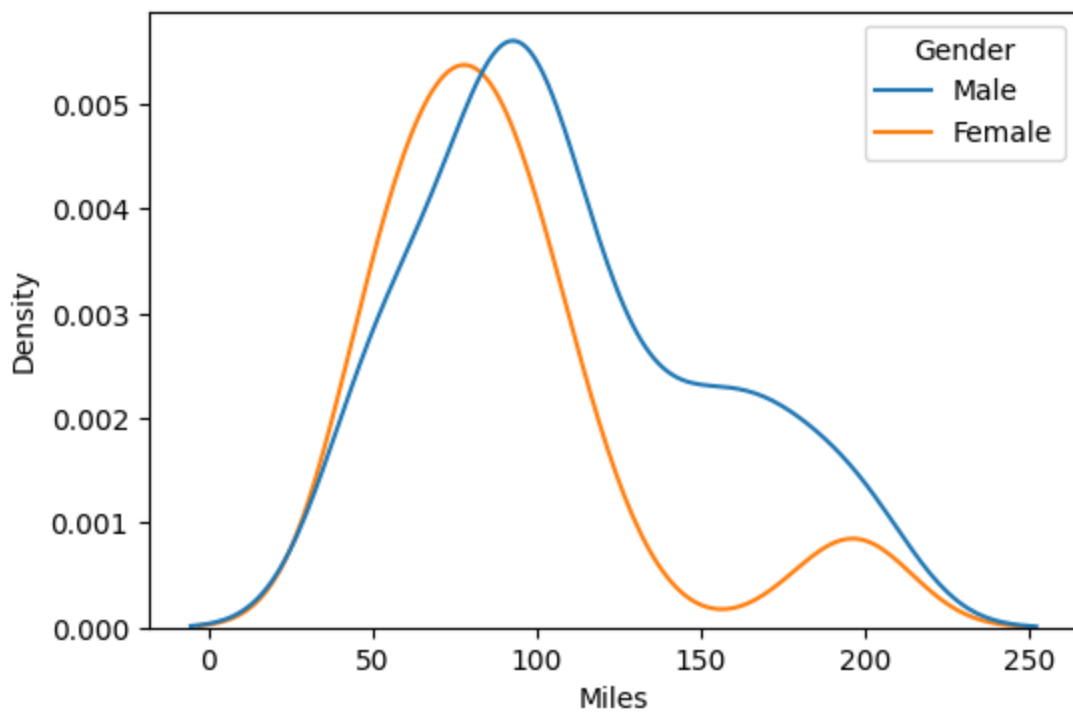
Business Insights:

Tailoring Marketing Strategies: Consider adjusting marketing strategies based on gender-specific usage patterns. For example:

- **For females:** Focus on features that resonate with their preferences.
- **For males:** Enhance engagement with male users.

```
In [67]: # Distance covered by each Gender among the customers
```

```
plt.figure(figsize=(6,4))
sns.kdeplot(data=df,x='Miles',hue='Gender')
plt.show()
```



Observation:

- Male customers plan on covering more distance on average each week
- Females, on the other hand, expect to cover shorter distances on average each week

Business Insights:

Product Development: Consider tailoring products or services based on these usage patterns:

- **For males:** Develop endurance-focused offerings or longer workout programs.
- **For females:** Create shorter, engaging workout sessions.

Promotional Strategies:

- **Design gender-specific promotions:** Challenges or rewards aligned with the observed average distances covered by each gender.
- Encourage both groups to achieve their fitness goals.

Using concept of averages to get better understanding for Numerical Columns:

```
In [68]: # Average usage of each product type by the customer
df.groupby('Product')['Usage'].mean()
```

```
Out[68]: Product
KP281    3.087500
KP481    3.066667
KP781    4.511250
Name: Usage, dtype: float64
```

- Mean usage for product KP281 is 3.08
- Mean usage for product KP481 is 3.06
- Mean usage for product KP781 is 4.5

```
In [69]: # Average Age of customer using each product
df.groupby('Product')['Age'].mean()
```

```
Out[69]: Product
KP281    28.427500
KP481    28.801667
KP781    28.828750
Name: Age, dtype: float64
```

- Mean Age of the customer who purchased product KP281 is 28.42
- Mean Age of the customer who purchased product KP481 is 28.80
- Mean Age of the customer who purchased product KP781 is 28.8

```
In [70]: # Average Education of customer using each product
df.groupby('Product')['Education'].mean()
```

```
Out[70]: Product
KP281    15.125000
KP481    15.183333
KP781    17.050000
Name: Education, dtype: float64
```

- Mean Education qualification of the customer who purchased product KP281 is 15 years
- Mean Education qualification of the customer who purchased product KP481 is 15 years
- Mean Education qualification of the customer who purchased product KP781 is 17 years

```
In [71]: # Average customer fitness rating for each product type purchased
df.groupby('Product')['Fitness'].mean()
```

```
Out[71]: Product
KP281    2.975000
KP481    2.916667
KP781    4.625000
Name: Fitness, dtype: float64
```

- Customer fitness mean for product KP281 is 2.96 approx 3
- Customer fitness mean for product KP481 is 2.90 approx 3
- Customer fitness mean for product KP781 is 4.62 approx 5

3.b.2. Find if there is any relationship between the continuous variables and the output variable in the data : Using Scatterplot

```
In [28]: # Set up the subplots
fig, axes = plt.subplots(3, 3, figsize=(15, 15))

# Scatterplot: Age vs. Income with Product as hue
sns.scatterplot(data=df, x='Age', y='Income', hue='Product', ax=axes[0, 0])
axes[0, 0].set_title('Age vs. Income')

# Scatterplot: Fitness vs. Usage with Product as hue
sns.scatterplot(data=df, x='Fitness', y='Usage', hue='Product', ax=axes[0, 1])
axes[0, 1].set_title('Fitness vs. Usage')

# Scatterplot: Miles vs. Usage with Product as hue
sns.scatterplot(data=df, x='Miles', y='Usage', hue='Product', ax=axes[0, 2])
axes[0, 2].set_title('Miles vs. Usage')

# Scatterplot: Income vs. Product Purchased with Product as hue
sns.scatterplot(data=df, x='Income', y='Product', hue='Product', marker='o', s=100, ax=axes[1, 0])
axes[1, 0].set_title('Income vs. Product Purchased')
```

```
# Scatterplot: Age vs. Miles with Product as hue
sns.scatterplot(data=df, x='Age', y='Miles', hue='Product', ax=axes[1, 1])
axes[1, 1].set_title('Age vs. Miles')

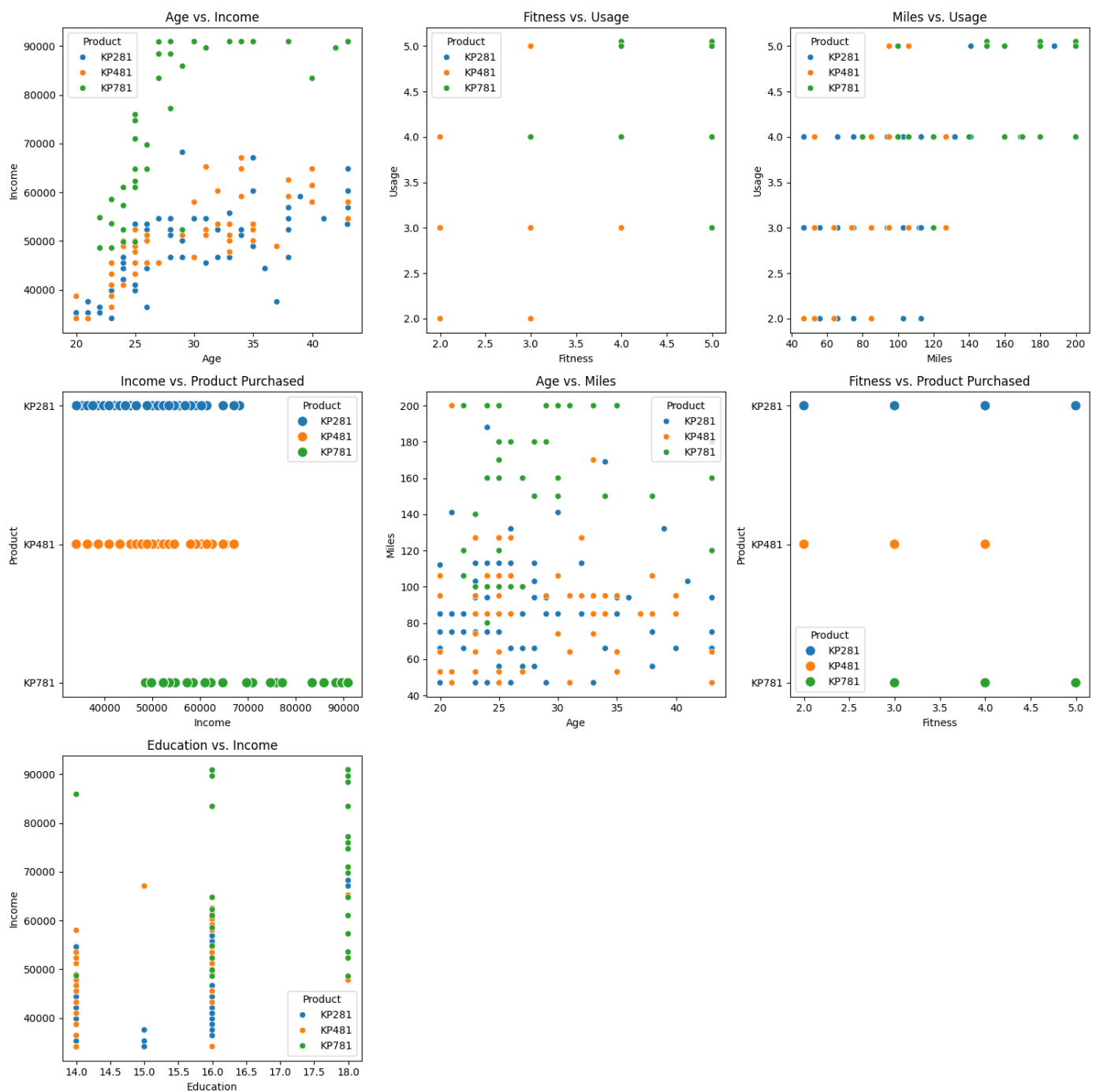
# Scatterplot: Fitness vs. Product Purchased with Product as hue
sns.scatterplot(data=df, x='Fitness', y='Product', hue='Product', marker='o', s=100, ax=
axes[1, 2].set_title('Fitness vs. Product Purchased')

# Scatterplot: Education vs. Income with Product as hue
sns.scatterplot(data=df, x='Education', y='Income', hue='Product', ax=axes[2, 0])
axes[2, 0].set_title('Education vs. Income')

# Hide the last subplot (empty)
axes[2, 1].axis('off')

# Hide the last subplot (empty)
axes[2, 2].axis('off')

# Adjust layout
plt.tight_layout()
plt.show()
```



Scatter Plot	Observations	Business Insights
Age vs. Income	- KP281 (Blue): Income ranges from \$40,000 to \$70,000 for ages 20 to 40. Concentration around \$45,000 to \$60,000. - KP481 (Orange): Income ranges from \$40,000 to \$75,000 for ages 20 to 40. More scattered, with a similar concentration to KP281. - KP781 (Green): Income ranges from 50,000 to 90,000 for ages 25 to 40. Higher concentration above 70,000.	- KP781: Appeals to higher-income individuals. Market towards affluent demographics, highlighting premium features. - KP481 & KP281: Similar income demographics. Focus on affordability and value for money.
Fitness vs. Usage	- KP281 (Blue): Usage around 3.0 to 4.0 times per week across fitness levels. - KP481 (Orange): Usage mostly at 3.0 times per week across fitness levels. - KP781 (Green): Usage around 4.0 to 5.0 times per week, with high consistency across fitness levels.	- KP781: Suitable for both novice and experienced users. Market highlighting versatility and high usage. - KP281 & KP481: KP281 shows moderate usage. KP481 requires improvements to increase engagement.
Miles vs. Usage	- KP281 (Blue): Usage around 3.0 to 4.0 times per week, miles ranging from 40 to 140. - KP481 (Orange): Usage around 2.0 to 3.0 times per week, miles ranging from 40 to 120. - KP781 (Green): Usage consistently around 4.0 to 5.0 times per week, miles ranging from 40 to 200. Higher miles indicate more intensive use.	- KP781: High user satisfaction, marketed for intensive workouts. - KP281: Balanced option for regular exercise. Market as reliable and value option. - KP481: Needs improvements to increase usage.

Observations for remaining features:

Product	Target Audience	Income Range	Age Range	Fitness Level	Marketing Strategy
KP281	Younger individuals, beginners, or those with lower to mid-income	\$40,000 - \$80,000	20-30	2 - 5	Highlight affordability, accessibility, and features for beginners
KP481	Middle-income customers with moderate fitness levels	\$50,000 - \$70,000	20-40	3	Emphasize value for money, durability, and suitability for regular users
KP781	Higher-income, older, and more fitness-conscious individuals	\$60,000 - \$90,000	20-40	4 - 5	Focus on advanced features, high performance, and the ability to support intense workouts

Observations for Education and Income features:

Scatter Plot	Observations	Business Insights
Education vs. Income	- KP281 (Blue): Concentration of income around \$45,000 to \$60,000 across education levels 14 to 18 years. - KP481 (Orange): Similar income distribution to KP281, with a few higher income outliers around 70,000. - KP781 (Green): Higher income distribution with concentrations above	- KP781: Appeals to more educated, higher-income individuals. Market as a premium product for educated professionals. - KP481 & KP281: Similar income demographics. Focus on value and reliability for users with varying education levels.

Summary of Business Insights

- **Product Differentiation:**
 - **KP781:** Suited for higher-income, more educated, high-usage consumers. Market as a premium product.
 - **KP281:** Balanced option for mid-income users. Emphasize reliability and value.
 - **KP481:** Needs product enhancements or new features. Focus marketing on improvements and upgrades.
- **Marketing Strategies:**
 - **KP781:** Focus on affluent, high-usage consumers highlighting premium features and benefits.
 - **KP281:** Emphasize affordability, reliability, and consistent performance for mid-income users.
 - **KP481:** Address lower usage through targeted improvements and highlight these changes in marketing campaigns.

3.c. Multivariate Analysis:

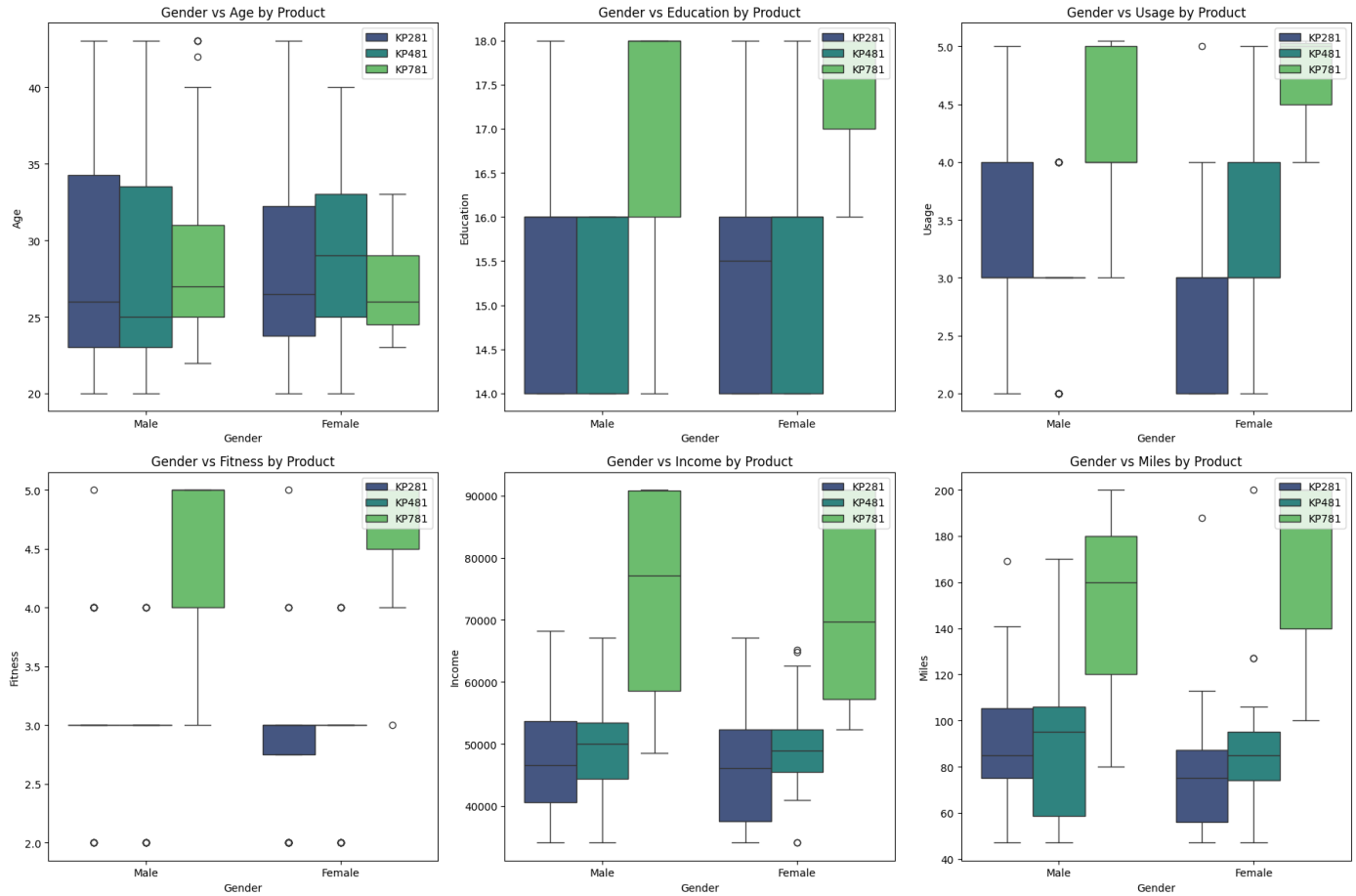
```
In [29]: numerical_columns = ['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']

# Plot size and layout
fig, axs = plt.subplots(2, 3, figsize=(18, 12))

# Flatten the axes array for easier iteration
axs = axs.flatten()

# Generate the boxplots
for i, column in enumerate(numerical_columns):
    sns.boxplot(data=df, x='Gender', y=column, hue='Product', ax=axs[i], palette='viridi
    axs[i].set_title(f'Gender vs {column} by Product')
    axs[i].legend(loc='upper right')

plt.tight_layout()
plt.show()
```



Category	Product	Observation	Insight
Age	KP281	- Males have a slightly younger median age than females.	Both genders have a wide age range for this product.
	KP481	- Median age for males is slightly higher than for females. - Less variation in age among females.	
	KP781	- Similar median ages for both genders. - Females have a smaller age range.	
Education	KP281	- Education levels are similar. - Males have a slightly higher median.	Education levels are fairly consistent across genders.
	KP481	- Almost identical education levels between genders.	
	KP781	- Males have a slightly higher median education level. - Females show a larger range.	
Usage	KP281	- Similar usage frequency between genders. - Males show a wider range.	Usage patterns are consistent across genders.
	KP481	- Males have a slightly higher median usage than females.	
	KP781	- Females use the product more frequently than males.	

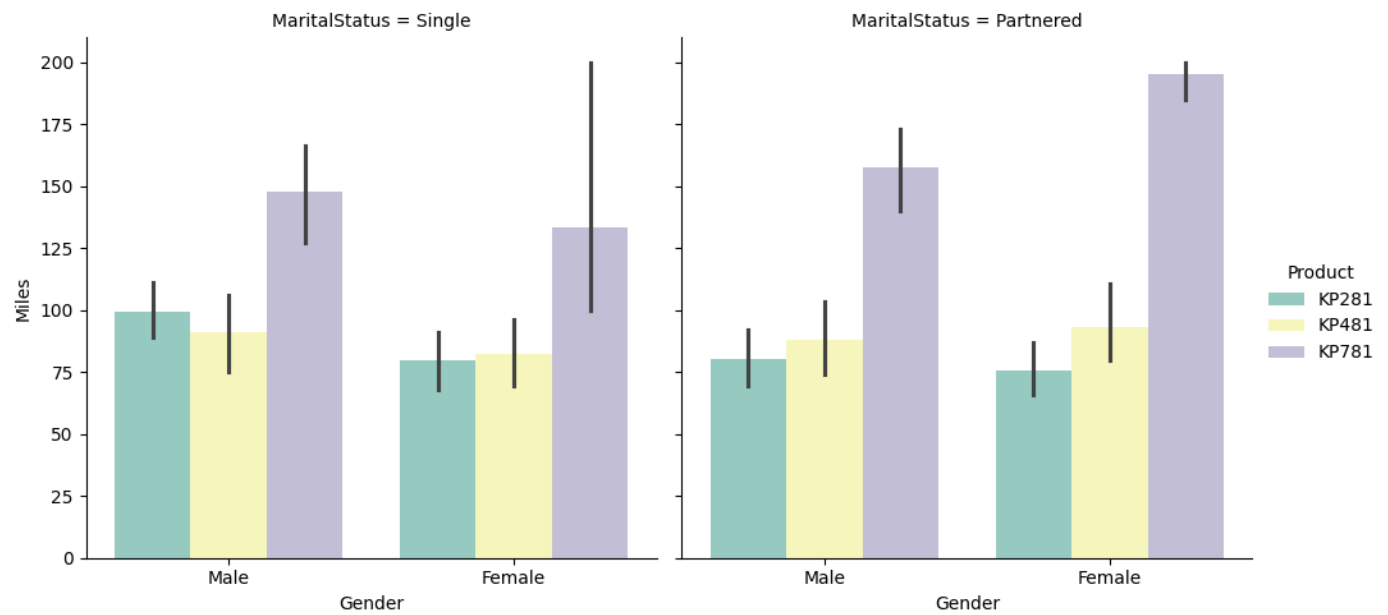
		- Higher median usage for females.	likely to choose the KP781 product.
Fitness	KP281	- Both genders have a median fitness level of 3.	Consistent fitness levels across genders.
	KP481	- Median fitness level of 3 for both genders. - Slightly more variation in males.	Fitness levels are similar across genders.
	KP781	- Females have a higher median fitness level than males.	Females perceive themselves to be in better shape.
Income	KP281	- Income levels slightly higher for males. - Similar range for both genders.	Males have slightly higher incomes.
	KP481	- Similar income distributions. - Females have a slightly higher median income.	Females with moderate incomes are likely to choose the KP481 product.
	KP781	- Females have significantly higher median income. - Larger range compared to males.	Potential marketing point: higher-income female customers are inclined towards the KP781 product.
Miles	KP281	- Males and females have similar median miles. - Males show slightly more variation.	Similar expected usage in terms of miles.
	KP481	- Similar distributions. - Males tend to run slightly more miles.	Consistent usage patterns across genders.
	KP781	- Females have a higher median number of miles than males. - Indicating higher expected usage.	Females expecting to run more miles per week are likely to choose the KP781 product.

What does it mean for Business ?

Category	Product	Business Insight
Age	KP281	- The KP281 attracts younger customers across both genders, suggesting it's perceived as an entry-level treadmill.
	KP481	- KP481 appeals to a slightly older demographic, particularly males, indicating it may be favored by more experienced users.
	KP781	- The KP781 appeals equally across ages and genders, indicating it's suitable for a wide audience looking for advanced features.
Education	KP281	- The KP281 is popular among customers with varying education levels, suggesting it's accessible and easy to use.
	KP481	- The KP481 is chosen by both genders with similar education levels, indicating it's seen as a balanced option.
	KP781	- More educated females tend to buy

		the KP781, implying it may be marketed as a premium, high-tech option appealing to educated consumers.
Usage	KP281	- The KP281 is used frequently by both genders, showing it's reliable for regular use.
	KP481	- The KP481's usage pattern indicates it may be ideal for slightly more serious users who require consistent performance.
	KP781	- Females planning to use the treadmill more frequently prefer the KP781, suggesting it's perceived as durable and reliable for heavy use.
Fitness	KP281	- Consistent fitness levels among KP281 users indicate it's suitable for maintaining basic fitness routines.
	KP481	- KP481's balanced fitness levels suggest it's ideal for moderate to advanced fitness enthusiasts.
	KP781	- Higher fitness levels among KP781 users, especially females, indicate it's preferred by those aiming for peak physical condition.
Income	KP281	- The KP281 is affordable for both genders, making it a budget-friendly choice for lower to moderate income groups.
	KP481	- KP481's similar income distribution indicates it's a mid-range option, appealing to a broad spectrum of incomes.
	KP781	- Higher incomes among KP781 buyers, particularly females, suggest it's marketed as a premium product for affluent customers.
Miles	KP281	- The KP281's users run a moderate number of miles, indicating it's effective for regular but not intensive use.
	KP481	- Similar miles run by KP481 users suggest it's reliable for consistent exercise routines.
	KP781	- Higher miles expected by KP781 users, especially females, show it's preferred for extensive use, highlighting its endurance and advanced features.

```
In [57]: # Miles covered in each product by gender and their marital status
sns.catplot(x='Gender',y='Miles',hue='Product',col='MaritalStatus',data=df,kind='bar',pa
plt.show())
```

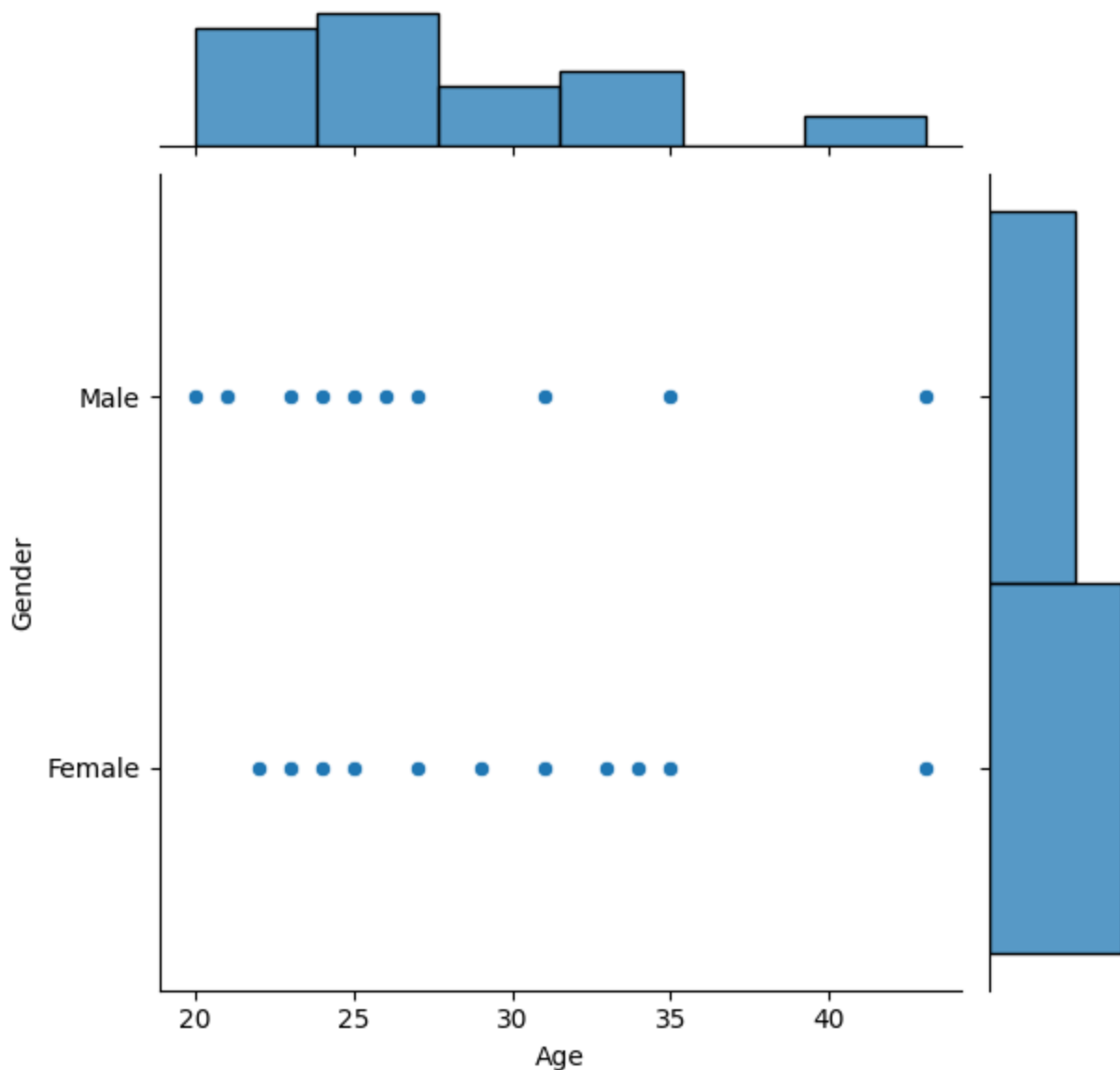


Observations:

- Female customers (both single and partnered) tend to run more miles than male customers.
- Among partnered customers, females run significantly more miles than males, especially for KP281 and KP781.
- KP781 has the highest usage overall, followed by KP281 and then KP481.

```
In [77]: # Scatterplot for customers Gender and Age who rated less than 2 in Fitness rating
plt.figure(figsize=(6,4))
sns.jointplot(x='Age',y='Gender',data=df[df.Fitness<3])
plt.show()
```

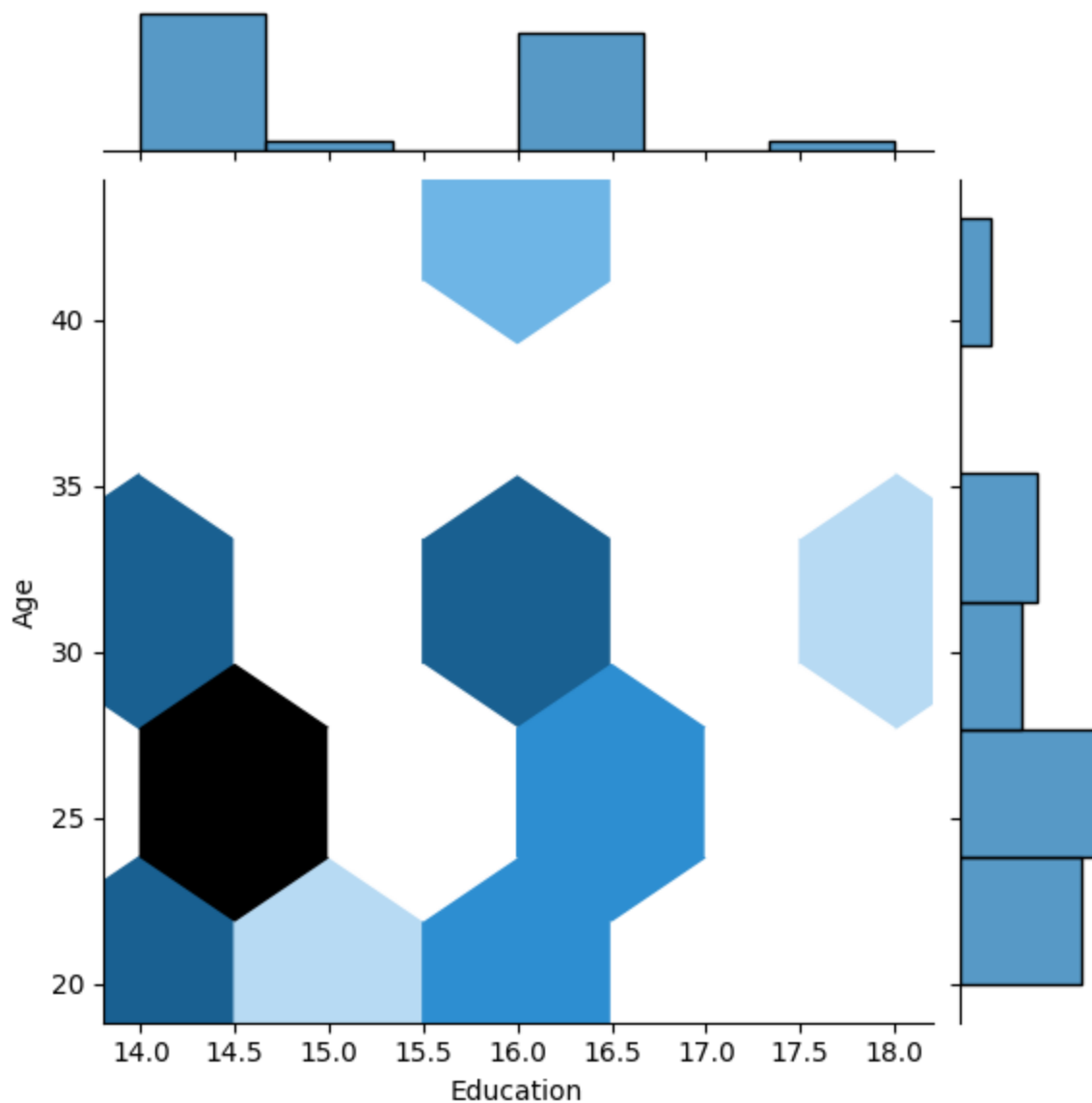
<Figure size 600x400 with 0 Axes>



Graph 1: Scatterplot for Gender and Age who rated less than 2 in Fitness rating

- **Customer Age and Gender Relationship:** This plot illustrates the relationship between the age and gender of customers who rated their fitness level as less than 2.
- **Gender Distribution:** The data shows that both males and females are present in the group of customers with low fitness ratings. However, there is a noticeable trend:
 - **For Females:** Younger women, especially those in their early twenties, are more likely to rate their fitness as less than 2.
 - **For Males:** Men in their twenties also show low fitness ratings, but there are a few older men (up to mid-forties) who rate their fitness similarly low.
- **Age Distribution:** The product appears to be less familiar or appealing to older women, as there are significantly fewer women above 35 with low fitness ratings compared to men. This could indicate a potential area for targeted marketing or improvement.

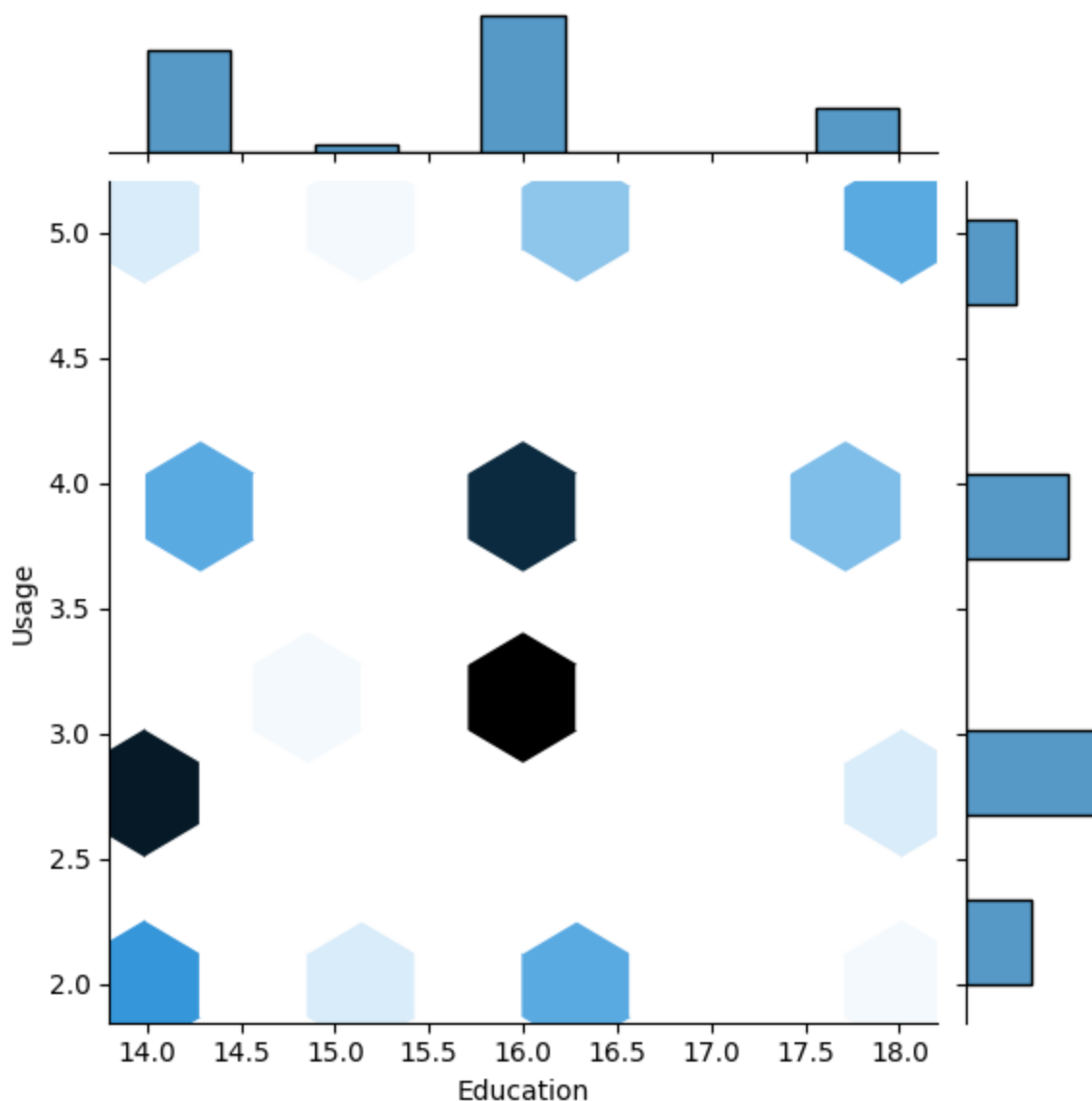
```
In [79]: # Hex Scatterplot for customers Education and Age who rated less than 2 in Fitness rating
sns.jointplot(x='Education',y='Age',kind='hex',data=df[df.Fitness<3])
plt.show()
```



Hex Scatterplot for Education and Age who rated less than 2 in Fitness rating

- **Customer Education and Age Relationship:** This hex scatterplot shows the relationship between the education level and age of customers who rated their fitness level as less than 2.
- **Age and Education Distribution:**
 - **Young Adults:** The highest concentration of low fitness ratings is among young adults aged 25-30 years with education levels around 14-16 years. This suggests that younger individuals with moderate education are more likely to rate their fitness poorly.
 - **Middle-Aged Customers:** There are a few middle-aged customers (35-40 years) with education levels between 15-16 years who also rate their fitness less than 2, but they are fewer compared to younger customers.
- **Educational Impact:**
 - **Moderate Education Levels:** Customers with education levels around 14-16 years are the most likely to report low fitness ratings.
 - **Higher Education Levels:** There are fewer customers with higher education levels (17-18 years) who rate their fitness as less than 2, indicating that higher education may correlate with higher fitness ratings or at least a perception of better fitness.

```
In [80]: sns.jointplot(x='Education', y='Usage', data=df, kind='hex')
plt.show()
```



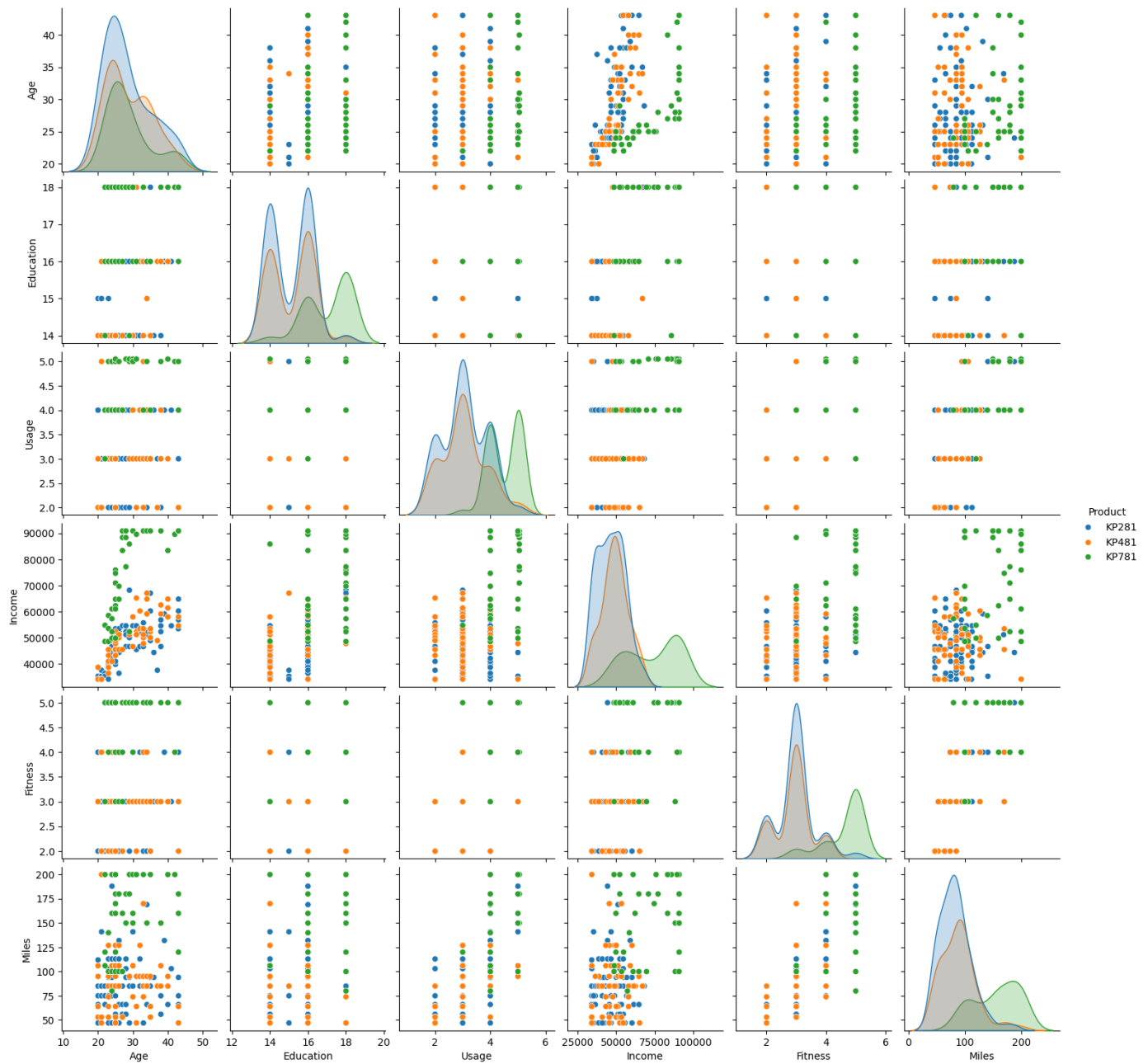
Graph 3: Hexbin Plot for Education and Usage

- **Customer Education and Usage Relationship:** This hexbin plot highlights the relationship between the education level of customers and their usage of the product.
- **Usage Patterns:**
 - **High Concentration:** The highest concentration of customers is seen at the intersection of 15-16 years of education and a usage level of around 3. This suggests that customers with a moderate education level are the most frequent users of the product.
 - **Moderate Usage:** Most customers with a variety of education levels (14-18 years) use the product moderately (usage level 3).
- **Distinct Groups:**
 - **Higher Education, High Usage:** Customers with higher education levels (around 17-18 years) tend to either use the product very frequently or very infrequently, indicating a polarized usage pattern.
 - **Lower Education, Variable Usage:** Those with lower education levels (14-15 years) also show a mix of usage patterns but tend to concentrate more around moderate usage.
- **Implications:** These insights can guide marketing strategies, suggesting that promotional efforts could be focused on individuals with 15-16 years of education to maximize engagement. Additionally, understanding the polarized usage among highly educated customers could help tailor features or services to better meet their needs.

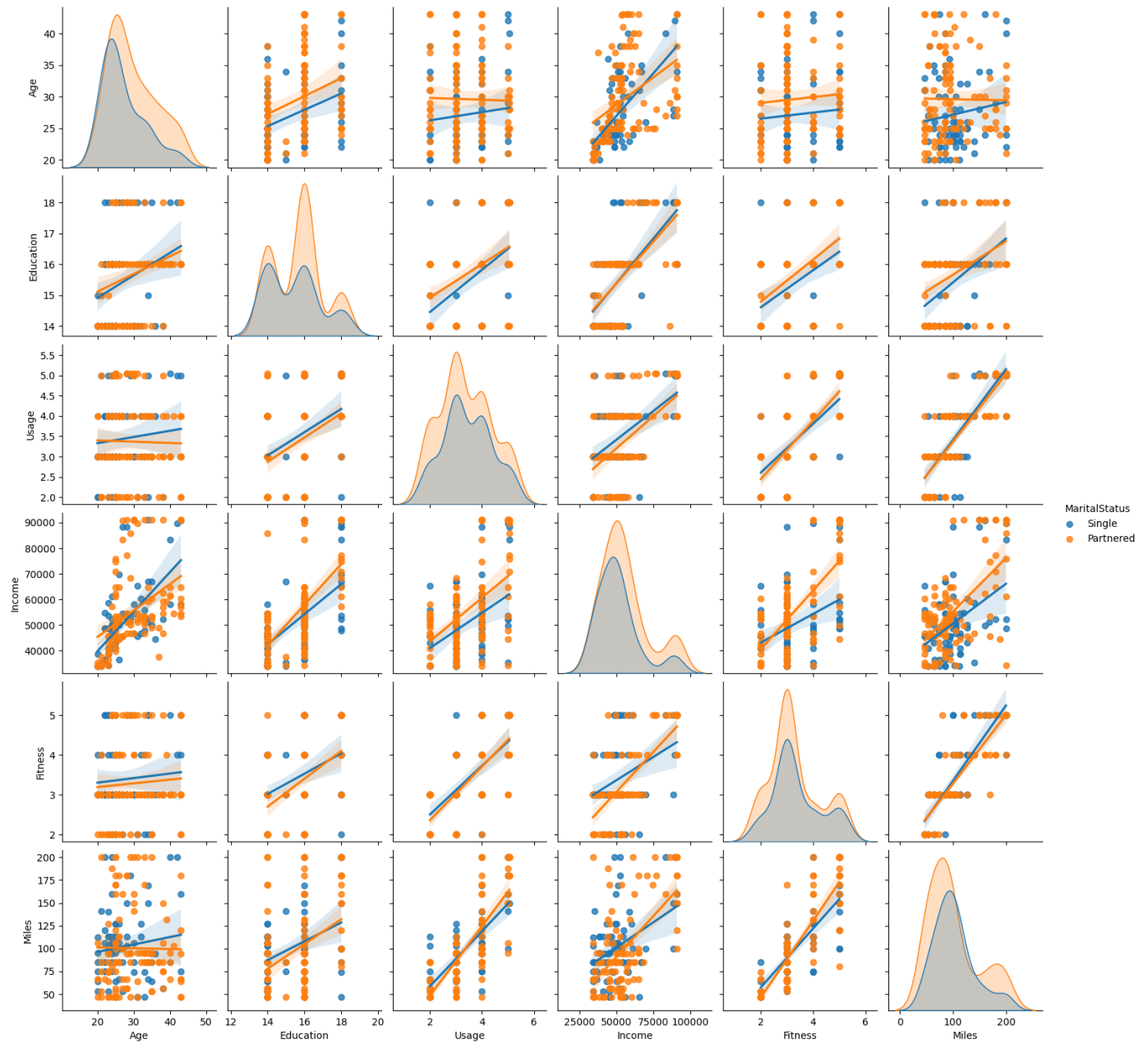
Using Pairplots

```
In [59]: # Create pair plots
sns.pairplot(df, hue='Product', diag_kind='kde',
             vars=['Age', 'Education', 'Usage', 'Income', 'Fitness', 'Miles'])

# Show the plot
plt.show()
```

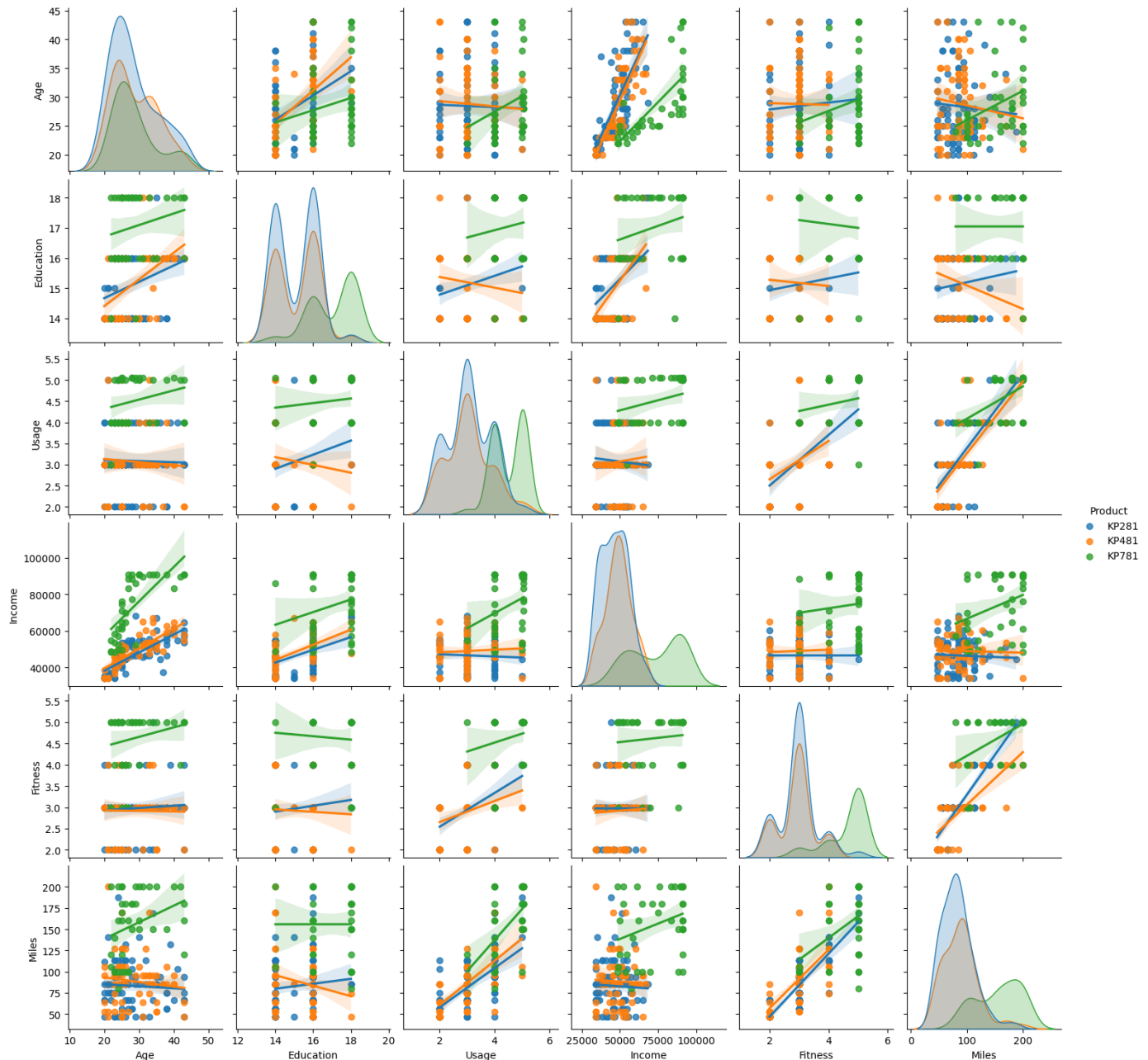


```
In [60]: # Marital Status - Pair Plot
sns.pairplot(df, hue='MaritalStatus', kind='reg',
             vars=['Age', 'Education', 'Usage', 'Income', 'Fitness', 'Miles'])
plt.show()
```



```
In [61]: # Product Analysis - Pair Plot
sns.pairplot(df, hue='Product', kind='reg',
              vars=['Age', 'Education', 'Usage', 'Income', 'Fitness', 'Miles'])
```

```
Out[61]: <seaborn.axisgrid.PairGrid at 0x7c1789061750>
```



Observations:

The pairplots coincide with the observations done so far .

Step 4: Compute Probabilities

4.a.Finding Marginal Probabilities:

Marginal probability is a straightforward way to see how popular each treadmill model is among all the treadmills sold.

```
In [31]: product_counts = pd.crosstab(index=df['Product'], columns='count', margins=True, margins_n

# Calculate the marginal probabilities manually ( without normalize)
# marginal_probabilities = product_counts / product_counts.sum()
```

```
# marginal_probabilities
```

```
product_counts
```

```
Out[31]:
```

	col_0	count	Total
	Product		
	KP281	80	80
	KP481	60	60
	KP781	40	40
	Total	180	180

```
In [32]:
```

```
# Create a crosstab to calculate the marginal probabilities
marginal_probabilities = pd.crosstab(index=df['Product'], columns='count', normalize=True)

marginal_probabilities
```

```
Out[32]:
```

	col_0	count
	Product	
	KP281	0.444444
	KP481	0.333333
	KP781	0.222222

Marginal Probability Interpretation for Treadmill Models

Treadmill Model	Marginal Probability	Explanation
KP281	0.444444	If you look at all the customers who bought a treadmill, about 44 out of every 100 of them chose the KP281 model. This means that KP281 is quite popular among the customers.
KP481	0.333333	Out of all the customers, roughly 33 out of every 100 bought the KP481 treadmill. This shows that KP481 is also a popular choice, though not as popular as KP281.
KP781	0.222222	When considering all the treadmill purchases, around 22 out of every 100 customers went for the KP781 model. This indicates that KP781 is the least chosen model compared to the other two.

Summary

- **KP281:** Almost half of the customers bought this model, showing it's the favorite.
- **KP481:** About one-third of the customers chose this treadmill, making it the second favorite.
- **KP781:** A little over one-fifth of the customers bought this model, indicating it's the least preferred but still significant.

What This Means

- **Popularity Insight:** Marginal probability helps us understand which treadmill models are most and least popular among customers.
- **Business Decisions:** Knowing these probabilities can help AeroFit focus its marketing and stock more effectively. For example, since the KP281 is the most popular, they might want to produce more of it or highlight it in advertisements. On the other hand, they might explore why KP781 is less popular and look for ways to improve its appeal or find a niche market that values its features.

4.b. Find the probability that the customer buys a product based on each column

```
In [33]: # Crosstab to calculate conditional probabilities
product_gender_prob = pd.crosstab(df['Product'], df['Gender'], normalize='index')
product_education_prob = pd.crosstab(df['Product'], df['Education'], normalize='index')
product_marital_prob = pd.crosstab(df['Product'], df['MaritalStatus'], normalize='index')
product_usage_prob = pd.crosstab(df['Product'], df['Usage'], normalize='index')
product_fitness_prob = pd.crosstab(df['Product'], df['Fitness'], normalize='index')
product_age_prob = pd.crosstab(df['Product'], df['Age'], normalize='index')
product_income_prob = pd.crosstab(df['Product'], df['Income'], normalize='index')
product_miles_prob = pd.crosstab(df['Product'], df['Miles'], normalize='index')

print("Probability of buying a product based on Gender:")
product_gender_prob
```

Probability of buying a product based on Gender:

Product	Gender	Female	Male
KP281		0.500000	0.500000
KP481		0.483333	0.516667
KP781		0.175000	0.825000

```
In [34]: df.columns
```

```
Out[34]: Index(['Product', 'Age', 'Gender', 'Education', 'MaritalStatus', 'Usage',
        'Fitness', 'Income', 'Miles'],
        dtype='object')
```

Insights Based on Probabilities of Buying a Product Based on Gender:

1. **KP281 and KP481 have balanced gender distribution:** Both the entry-level (KP281) and mid-level (KP481) treadmills show an almost equal probability of purchase between male and female customers, each at around 50%.
2. **KP781 has a significant gender preference:** The advanced treadmill (KP781) shows a considerable difference in purchasing behavior between male and female customers. The probability of purchase for males is significantly higher at 82.5%, compared to only 17.5% for females.
3. **Potential market segmentation opportunity:** Given the gender disparity in purchasing behavior for KP781, there might be an opportunity for AeroFit to explore targeted marketing campaigns or product features that appeal more to female customers to tap into this segment of the market and potentially increase sales for this product.

```
In [35]: print("\nProbability of buying a product based on Education:")
product_education_prob
```

Probability of buying a product based on Education:

Product	Education	14	15	16	18
	Product				
KP281		0.437500	0.050000	0.487500	0.025000
KP481		0.433333	0.016667	0.516667	0.033333
KP781		0.050000	0.000000	0.375000	0.575000

Insights Based on Probabilities of Buying a Product Based on Education:

- Education Level 16 shows highest probability across products:** Across all three treadmill products, customers with an education level of 16 have the highest probability of purchase. For KP281, KP481, and KP781, the probabilities are 48.75%, 51.67%, and 37.5% respectively.
- Education Level 14 and 15 show moderate probabilities:** Customers with education levels of 14 and 15 show moderate probabilities of purchasing the treadmills. However, these probabilities vary across products. For example, for KP281, the probabilities are 43.75% and 5%, respectively.
- Education Level 18 shows significant preference for KP781:** Customers with an education level of 18 show a significant preference for the advanced treadmill (KP781), with a probability of purchase at 57.5%. This suggests that individuals with higher education levels might be more inclined towards purchasing premium fitness equipment.

```
In [36]: print("\nProbability of buying a product based on Marital Status:")
product_marital_prob
```

Probability of buying a product based on Marital Status:

Product	MaritalStatus	Partnered	Single
	Product		
KP281		0.600	0.400
KP481		0.600	0.400
KP781		0.575	0.425

Insights Based on Probabilities of Buying a Product Based on Marital Status:

- KP281 and KP481 have similar probabilities across marital statuses:** Both the entry-level (KP281) and mid-level (KP481) treadmills show similar probabilities of purchase between customers who are partnered and single, each at 60% and 40% respectively.
- KP781 shows a slight preference for partnered customers:** The advanced treadmill (KP781) shows a slightly higher probability of purchase for partnered customers at 57.5% compared to 42.5% for single customers. This indicates that partnered individuals might be more inclined towards purchasing the premium treadmill compared to single individuals.

3. **Consideration for marketing strategies:** Given the differences in purchasing behavior based on marital status, AeroFit could tailor marketing strategies or promotions targeting partnered individuals differently from those targeting single individuals to maximize sales for each product.

```
In [37]: print("\nProbability of buying a product based on Usage:")
product_usage_prob
```

Probability of buying a product based on Usage:

Usage	2.00	3.00	4.00	5.00	5.05
Product					
KP281	0.237500	0.462500	0.275	0.025	0.000
KP481	0.233333	0.516667	0.200	0.050	0.000
KP781	0.000000	0.025000	0.450	0.300	0.225

Insights Based on Probabilities of Buying a Product Based on Usage:

1. **KP281 and KP481 show highest probabilities at Usage 3.00:** Both the entry-level (KP281) and mid-level (KP481) treadmills have the highest probabilities of purchase when the expected usage is 3 times per week, at 46.25% and 51.67% respectively.
2. **Usage 4.00 and 5.00 also have notable probabilities across products:** While the probabilities decrease compared to Usage 3.00, there are still significant probabilities of purchase at Usage 4.00 and 5.00 for all three treadmill products.
3. **KP781 shows significant preference for higher usage frequencies:** The advanced treadmill (KP781) shows a preference for higher usage frequencies compared to KP281 and KP481. Notably, at Usage 5.00, KP781 has a probability of purchase at 30%, whereas KP281 and KP481 have probabilities close to 0%.
4. **Consideration for product positioning and marketing:** Understanding the relationship between expected usage and probability of purchase can inform product positioning and marketing strategies. For instance, emphasizing durability, advanced features, or long-term value for higher usage frequencies could be effective for promoting the KP781 treadmill.

```
In [38]: print("\nProbability of buying a product based on Fitness:")
product_fitness_prob
```

Probability of buying a product based on Fitness:

Fitness	2	3	4	5
Product				
KP281	0.187500	0.675	0.112500	0.025
KP481	0.216667	0.650	0.133333	0.000
KP781	0.000000	0.100	0.175000	0.725

Insights Based on Probabilities of Buying a Product Based on Fitness:

1. **KP281 and KP481 show highest probabilities at Fitness 3:** Both the entry-level (KP281) and mid-level (KP481) treadmills have the highest probabilities of purchase when the self-rated fitness level is 3, at 67.5% for KP281 and 65% for KP481.
2. **Fitness level 4 also has notable probabilities across products:** While the probabilities decrease compared to Fitness 3, there are still significant probabilities of purchase at Fitness 4 for all three treadmill products.
3. **KP781 shows a significant preference for higher fitness levels:** The advanced treadmill (KP781) shows a preference for higher fitness levels compared to KP281 and KP481. At Fitness 5, KP781 has the highest probability of purchase at 72.5%.
4. **Consideration for product features and marketing:** Understanding the relationship between self-rated fitness level and probability of purchase can inform product features and marketing strategies. Emphasizing advanced training programs, performance tracking features, or athlete endorsements could be effective for promoting the KP781 treadmill to customers with higher fitness levels.

```
In [39]: print("\nProbability of buying a product based on Age:")
product_age_prob
```

Probability of buying a product based on Age:

```
Out[39]: Age 20.0021.0022.0023.0024.0025.0026.0027.0028.0029.00 ... 35.0036.0037.0038.0039.0040.0041.0042.0043.0043.05
Product
KP281 0.075000 0.0500 0.1000 0.0625 0.0875 0.0875 0.0375 0.0750 0.0375 0.0375 0.0125 0.0125 0.0500 0.0125 0.0125 0.0000 0.0125 0.0500
KP481 0.066667 0.0000 0.116667 0.0750 0.183333 0.0500 0.166667 0.0000 0.16667 0.0000 0.0500 0.0000 0.0000 0.0000 0.0000 0.033333
KP781 0.000000 0.0750 0.0750 0.0000 0.1750 0.0500 0.0750 0.0000 0.0750 0.0500 0.0000 0.0000 0.0250 0.0000 0.0250 0.0000 0.0750
```

3 rows × 25 columns

Product	20	21	22	23	24	25	26	27	28	29	...	35	36	37	38	39	40	41	42	43	43.05
KP281	7.50%	5.00%	10.00%	6.25%	8.75%	8.75%	3.75%	7.50%	3.75%	...	3.75%	1.25%	1.25%	5.00%	1.25%	1.25%	1.25%	0.00%	1.25%	5.00%	
KP481	6.67%	0.00%	11.67%	7.50%	18.33%	5.00%	1.67%	0.00%	1.67%	...	6.67%	0.00%	1.67%	3.33%	0.00%	5.00%	0.00%	0.00%	0.00%	3.33%	
KP781	0.00%	7.50%	7.50%	0.00%	17.50%	5.00%	7.50%	7.50%	5.00%	...	2.50%	0.00%	0.00%	2.50%	0.00%	2.50%	0.00%	2.50%	0.00%	7.50%	

Dividing the ages in to bins will provide better understanding:

```
In [40]: # Define age bins
age_bins = [20, 25, 30, 35, 40, 45, 50, 55, 60]

# Create labels for age bins
age_labels = ['20-24', '25-29', '30-34', '35-39', '40-44', '45-49', '50-54', '55-59']

# Add 'Age_Bin' column to the dataframe
df['Age_Bin'] = pd.cut(df['Age'], bins=age_bins, labels=age_labels, right=False)

# Crosstab to calculate probabilities based on age bins
product_age_prob = (pd.crosstab(df['Product'], df['Age_Bin'], normalize='index') * 100).

print("Probability of buying a product based on Age Bins:")
product_age_prob
```

Probability of buying a product based on Age Bins:

```
Out[40]:
```

Age_Bin	20-24	25-29	30-34	35-39	40-44
---------	-------	-------	-------	-------	-------

Product					
KP281	33.75%	32.5%	12.5%	12.5%	8.75%
KP481	28.33%	26.67%	25.0%	11.67%	8.33%
KP781	25.0%	42.5%	15.0%	5.0%	12.5%

Insights Based on Probabilities of Buying a Product Based on Age:

1. **KP281 and KP481:** These treadmills have higher probabilities of purchase among younger customers, particularly those in their early twenties.
 - **Ages 23 and 25 have significant purchase seen for KP481**
2. **KP781:** While also appealing to younger demographics, this treadmill shows a broader age distribution of probabilities, indicating its appeal to a wider age range, including customers in their late twenties and thirties.
 - **Ages 25 have significant purchase seen for KP781**
3. **Marketing Implications:** Targeting younger customers with promotions emphasizing affordability and trendy features could be effective for KP281 and KP481. For KP781, highlighting advanced technology and performance benefits may attract a broader demographic, including older age groups.

```
In [41]: # Dividing people in income groups would give a better understanding

# Define income bins
income_bins = [30000, 40000, 50000, 60000, 70000, 80000, 90000, 100000]

# Create labels for income bins
income_labels = ['30k-40k', '40k-50k', '50k-60k', '60k-70k', '70k-80k', '80k-90k', '90k-100k']

# Add 'Income_Bin' column to the dataframe
df['Income_Bin'] = pd.cut(df['Income'], bins=income_bins, labels=income_labels, right=False)

# Crosstab to calculate probabilities based on income bins
product_income_bin_prob = pd.crosstab(df['Product'], df['Income_Bin'], normalize='index')

print("Probability of buying a product based on Income Bins:")
product_income_bin_prob
```

Probability of buying a product based on Income Bins:

	Income_Bin	30k-40k	40k-50k	50k-60k	60k-70k	70k-80k	80k-90k	90k-100k
Product								
KP281		0.2875	0.3125	0.325000	0.075000	0.0	0.000	0.0
KP481		0.1500	0.3500	0.383333	0.116667	0.0	0.000	0.0
KP781		0.0000	0.1250	0.150000	0.150000	0.1	0.175	0.3

For better understanding showing probabilities as percentage

```
In [42]: # Crosstab to calculate probabilities based on income bins and display percentages
product_income_bin_prob = (pd.crosstab(df['Product'], df['Income_Bin'], normalize='index') * 100).round(2)

print("Probability of buying a product based on Income Bins:")
product_income_bin_prob
```

Probability of buying a product based on Income Bins:

Out[42]:

Income_Bin	30k-40k	40k-50k	50k-60k	60k-70k	70k-80k	80k-90k	90k-100k
Product							
KP281	28.75%	31.25%	32.5%	7.5%	0.0%	0.0%	0.0%
KP481	15.0%	35.0%	38.33%	11.67%	0.0%	0.0%	0.0%
KP781	0.0%	12.5%	15.0%	15.0%	10.0%	17.5%	30.0%

Insights Based on Probabilities of Buying a Product Based on Income Bins:

1. **KP281:** The majority of customers in the 40k-50k income bracket (31.25%) and 50k-60k income bracket (32.5%) prefer the entry-level treadmill (KP281). There's a significant drop in preference beyond the 50k-60k income range.
2. **KP481:** Customers with higher incomes show a preference for the mid-level treadmill (KP481). The highest preference is seen in the 50k-60k income bracket (38.33%), followed by the 40k-50k income bracket (35.0%).
3. **KP781:** The advanced treadmill (KP781) appeals to customers with higher incomes. The preference gradually increases with income, with the highest preference seen in the 90k-100k income bracket (30.0%). Notably, there's a substantial preference in the 80k-90k income bracket (17.5%) as well.

In [43]:

```
# Define miles bins
miles_bins = [0, 50, 100, 150, 200, 250]

# Create labels for miles bins
miles_labels = ['0-49', '50-99', '100-149', '150-199', '200-249']

# Add 'Miles_Bin' column to the dataframe
df['Miles_Bin'] = pd.cut(df['Miles'], bins=miles_bins, labels=miles_labels, right=False)

# Crosstab to calculate probabilities based on miles bins
product_miles_prob = (pd.crosstab(df['Product'], df['Miles_Bin'], normalize='index') * 100)

print("\nProbability of buying a product based on Miles:")
product_miles_prob
```

Probability of buying a product based on Miles:

Out[43]:

Miles_Bin	0-49	50-99	100-149	150-199	200-249
Product					
KP281	15.0%	62.5%	20.0%	2.5%	0.0%
KP481	8.33%	65.0%	21.67%	3.33%	1.67%
KP781	0.0%	2.5%	30.0%	40.0%	27.5%

Insights Based on Probabilities of Buying a Product Based on Miles:

1. **KP281:** Customers who expect to cover distances between **50 to 99 miles per week** show the highest preference for the entry-level treadmill (KP281), with **62.5%** probability. This suggests that this treadmill model is favored by those seeking moderate levels of usage.

2. **KP481**: The mid-level treadmill (KP481) also has a high preference among customers expecting to cover distances between **50 to 99 miles per week**, with a probability of **65.0%**. It maintains a relatively consistent preference across other mile ranges as well.
3. **KP781**: The advanced treadmill (KP781) shows an interesting trend where as the expected weekly distance increases, the preference for this treadmill also increases. Notably, it has the highest preference (**40.0%**) among customers expecting to cover distances between **150 to 199 miles per week**, indicating its suitability for more serious runners or fitness enthusiasts.

These insights suggest that the preference for each treadmill model varies based on the expected weekly distance, with different models catering to different usage levels and fitness goals.

4.c. Find the conditional probability that an event occurs given that another event has occurred.

(Example: given that a customer is female, what is the probability she'll purchase a KP481)

```
In [44]: # 1. Conditional probability of purchasing a product given Female/Male i.e Gender:
pd.crosstab(index=df["Product"], columns=df["Gender"], margins=True, margins_name="Total")
```

Out[44]:

Gender	Female	Male	Total
Product			
KP281	40	40	80
KP481	29	31	60
KP781	7	33	40
Total	76	104	180

For calculating conditional probability: Example : given that a customer is female, what is the probability she'll purchase a K2481

$$\begin{aligned}
 P(\text{KP281} \mid \text{Female}) &= P(\text{KP281} \cap \text{Female}) / P(\text{Female}) \\
 &= 40/76 \\
 &\approx 0.5263
 \end{aligned}$$

Same can be achieved using **normalize = "columns"**

```
In [45]: pd.crosstab(index=df["Product"], columns=df["Gender"], margins=True, margins_name="Total", normalize="columns")
```

Out[45]:

Gender	Female	Male	Total
Product			
KP281	0.526316	0.384615	0.444444
KP481	0.381579	0.298077	0.333333
KP781	0.092105	0.317308	0.222222

Conditional Probability	Mathematical Representation	Probability
Given that a customer is female, the probability she'll purchase a KP281	$P(\text{KP281} \mid \text{Female})$	52.63%
Given that a customer is female, the	$P(\text{KP481} \mid \text{Female})$	38.16%

probability she'll purchase a KP481		
Given that a customer is female, the probability she'll purchase a KP781	P(KP781 Female)	9.21%
Given that a customer is male, the probability he'll purchase a KP281	P(KP281 Male)	38.46%
Given that a customer is male, the probability he'll purchase a KP481	P(KP481 Male)	29.81%
Given that a customer is male, the probability he'll purchase a KP781	P(KP781 Male)	31.73%

```
In [46]: # 2.Conditional probability of purchasing a product given the age group:
pd.crosstab(index=df["Product"], columns=df["Age_Bin"], margins=True, margins_name="Total")
```

```
Out[46]:
```

Age_Bin	20-24	25-29	30-34	35-39	40-44	Total
Product						
KP281	0.500000	0.440678	0.322581	0.526316	0.411765	0.444444
KP481	0.314815	0.271186	0.483871	0.368421	0.294118	0.333333
KP781	0.185185	0.288136	0.193548	0.105263	0.294118	0.222222

Conditional Probability	Mathematical Representation	Probability
Given that a customer is in the age group 20-24, the probability	P(KP281 Age_Bin 20-24)	50.00%
they'll purchase KP281/KP481/KP781	P(KP481 Age_Bin 20-24)	31.48%
	P(KP781 Age_Bin 20-24)	18.52%
Given that a customer is in the age group 25-29, the probability	P(KP281 Age_Bin 25-29)	44.07%
they'll purchase KP281/KP481/KP781	P(KP481 Age_Bin 25-29)	27.12%
	P(KP781 Age_Bin 25-29)	28.81%
Given that a customer is in the age group 30-34, the probability	P(KP281 Age_Bin 30-34)	32.26%
they'll purchase KP281/KP481/KP781	P(KP481 Age_Bin 30-34)	48.39%
	P(KP781 Age_Bin 30-34)	19.35%
Given that a customer is in the age group 35-39, the probability	P(KP281 Age_Bin 35-39)	52.63%
they'll purchase KP281/KP481/KP781	P(KP481 Age_Bin 35-39)	36.84%
	P(KP781 Age_Bin 35-39)	10.53%
Given that a customer is in the age group 40-44, the probability	P(KP281 Age_Bin 40-44)	41.18%
they'll purchase KP281/KP481/KP781	P(KP481 Age_Bin 40-44)	29.41%
	P(KP781 Age_Bin 40-44)	29.41%

```
In [47]: # 3.Conditional probability of purchasing a product given the education level:
pd.crosstab(index=df["Product"], columns=df["Education"], margins=True, margins_name="Total")
```

```
Out[47]:
```

Education	14	15	16	18	Total
-----------	----	----	----	----	-------

Product					
KP281	0.555556	0.8	0.458824	0.074074	0.444444
KP481	0.412698	0.2	0.364706	0.074074	0.333333
KP781	0.031746	0.0	0.176471	0.851852	0.222222

Conditional Probability	Mathematical Representation	Probability
Given that a customer is in Education group 14, the probability	$P(\text{KP281} \mid \text{Education 14})$	55.56%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Education 14})$	41.27%
	$P(\text{KP781} \mid \text{Education 14})$	3.17%
Given that a customer is in Education group 15, the probability	$P(\text{KP281} \mid \text{Education 15})$	80.00%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Education 15})$	20.00%
	$P(\text{KP781} \mid \text{Education 15})$	0.00%
Given that a customer is in Education group 16, the probability	$P(\text{KP281} \mid \text{Education 16})$	45.88%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Education 16})$	36.47%
	$P(\text{KP781} \mid \text{Education 16})$	17.65%
Given that a customer is in Education group 18, the probability	$P(\text{KP281} \mid \text{Education 18})$	7.41%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Education 18})$	7.41%
	$P(\text{KP781} \mid \text{Education 18})$	85.19%
Given that a customer is in any Education group, the probability	$P(\text{KP281} \mid \text{Any Education})$	44.44%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Any Education})$	33.33%
	$P(\text{KP781} \mid \text{Any Education})$	22.22%

In [48]:

4.Conditional probability of purchasing a product given the marital status:
pd.crosstab(index=df["Product"], columns=df["MaritalStatus"], margins=True, margins_name

Out[48]:

MaritalStatus		Partnered	Single	Total
Product				
KP281		0.448598	0.438356	0.444444
KP481		0.336449	0.328767	0.333333
KP781		0.214953	0.232877	0.222222

Conditional Probability	Mathematical Representation	Probability
Given that a customer is Partnered, the probability	$P(\text{KP281} \mid \text{Partnered})$	44.86%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Partnered})$	33.64%
	$P(\text{KP781} \mid \text{Partnered})$	21.50%
Given that a customer is Single, the probability	$P(\text{KP281} \mid \text{Single})$	43.84%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Single})$	32.88%

In [49]: *# 5.Conditional probability of purchasing a product given the usage frequency:*

```
pd.crosstab(index=df["Product"], columns=df["Usage"], margins=True, margins_name="Total")
```

Out[49]:

Usage	2.0	3.0	4.0	5.0	5.049999999999983	Total
Product						
KP281	0.575758	0.536232	0.423077	0.117647	0.0	0.444444
KP481	0.424242	0.449275	0.230769	0.176471	0.0	0.333333
KP781	0.000000	0.014493	0.346154	0.705882	1.0	0.222222

Conditional Probability	Mathematical Representation	Probability
Given that a customer has Usage 2.0, the probability	$P(\text{KP281} \mid \text{Usage 2.0})$	57.58%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Usage 2.0})$	42.42%
	$P(\text{KP781} \mid \text{Usage 2.0})$	0.00%
Given that a customer has Usage 3.0, the probability	$P(\text{KP281} \mid \text{Usage 3.0})$	53.62%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Usage 3.0})$	44.93%
	$P(\text{KP781} \mid \text{Usage 3.0})$	1.45%
Given that a customer has Usage 4.0, the probability	$P(\text{KP281} \mid \text{Usage 4.0})$	42.31%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Usage 4.0})$	23.08%
	$P(\text{KP781} \mid \text{Usage 4.0})$	34.62%
Given that a customer has Usage 5.0, the probability	$P(\text{KP281} \mid \text{Usage 5.0})$	11.76%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Usage 5.0})$	17.65%
	$P(\text{KP781} \mid \text{Usage 5.0})$	70.59%
Given that a customer has Usage 5.049999999999983, the probability	$P(\text{KP281} \mid \text{Usage 5.049999999999983})$	0.00%
they'll purchase KP281/KP481/KP781	$P(\text{KP481} \mid \text{Usage 5.049999999999983})$	0.00%
	$P(\text{KP781} \mid \text{Usage 5.049999999999983})$	100.00%

In [50]: *# 6.Conditional probability of purchasing a product given the fitness level:*

```
pd.crosstab(index=df["Product"], columns=df["Fitness"], margins=True, margins_name="Total")
```

Out[50]:

Fitness	2	3	4	5	Total
Product					
KP281	0.535714	0.556701	0.375000	0.064516	0.444444
KP481	0.464286	0.402062	0.333333	0.000000	0.333333
KP781	0.000000	0.041237	0.291667	0.935484	0.222222

Conditional Probability	Mathematical Representation	Probability
Given that a customer has Fitness 2, the probability	$P(\text{KP281} \mid \text{Fitness 2})$	53.57%

they'll purchase KP281/KP481/KP781	P(KP481 Fitness 2)	46.43%
	P(KP781 Fitness 2)	0.00%
Given that a customer has Fitness 3, the probability	P(KP281 Fitness 3)	55.67%
they'll purchase KP281/KP481/KP781	P(KP481 Fitness 3)	40.21%
	P(KP781 Fitness 3)	4.12%
Given that a customer has Fitness 4, the probability	P(KP281 Fitness 4)	37.50%
they'll purchase KP281/KP481/KP781	P(KP481 Fitness 4)	33.33%
	P(KP781 Fitness 4)	29.17%
Given that a customer has Fitness 5, the probability	P(KP281 Fitness 5)	6.45%
they'll purchase KP281/KP481/KP781	P(KP481 Fitness 5)	0.00%
	P(KP781 Fitness 5)	93.55%

In [51]:

```
# 7.Conditional probability of purchasing a product given the income range:

pd.crosstab(index=df["Product"], columns=df["Income_Bin"], margins=True, margins_name="T
```

Out[51]:

Income_Bin	30k-40k	40k-50k	50k-60k	60k-70k	70k-80k	80k-90k	90k-100k	Total
Product								
KP281	0.71875	0.490196	0.472727	0.315789	0.0	0.0	0.0	0.444444
KP481	0.28125	0.411765	0.418182	0.368421	0.0	0.0	0.0	0.333333
KP781	0.00000	0.098039	0.109091	0.315789	1.0	1.0	1.0	0.222222

Conditional Probability	Mathematical Representation	Probability
Given that a customer has Income 30k-40k, the probability	P(KP281 Income 30k-40k)	71.88%
they'll purchase KP281/KP481/KP781	P(KP481 Income 30k-40k)	28.12%
	P(KP781 Income 30k-40k)	0.00%
Given that a customer has Income 40k-50k, the probability	P(KP281 Income 40k-50k)	49.02%
they'll purchase KP281/KP481/KP781	P(KP481 Income 40k-50k)	41.18%
	P(KP781 Income 40k-50k)	9.80%
Given that a customer has Income 50k-60k, the probability	P(KP281 Income 50k-60k)	47.27%
they'll purchase KP281/KP481/KP781	P(KP481 Income 50k-60k)	41.82%
	P(KP781 Income 50k-60k)	10.91%
Given that a customer has Income 60k-70k, the probability	P(KP281 Income 60k-70k)	31.58%
they'll purchase KP281/KP481/KP781	P(KP481 Income 60k-70k)	36.84%
	P(KP781 Income 60k-70k)	31.58%
Given that a customer has Income 70k-80k, the probability	P(KP281 Income 70k-80k)	0.00%
they'll purchase KP281/KP481/KP781	P(KP481 Income 70k-80k)	0.00%
	P(KP781 Income 70k-80k)	100.00%

Given that a customer has Income 80k-90k, the probability	P(KP281 Income 80k-90k)	0.00%
they'll purchase KP281/KP481/KP781	P(KP481 Income 80k-90k)	0.00%
	P(KP781 Income 80k-90k)	100.00%
Given that a customer has Income 90k-100k, the probability	P(KP281 Income 90k-100k)	0.00%
they'll purchase KP281/KP481/KP781	P(KP481 Income 90k-100k)	0.00%
	P(KP781 Income 90k-100k)	100.00%

```
In [52]: # 8.Conditional probability of purchasing a product given the expected weekly miles:
pd.crosstab(index=df["Product"], columns=df["Miles_Bin"], margins=True, margins_name="To
```

```
Out[52]:
```

Miles_Bin	0-49	50-99	100-149	150-199	200-249	Total
Product						
KP281	0.705882	0.555556	0.390244	0.1	0.000000	0.444444
KP481	0.294118	0.433333	0.317073	0.1	0.083333	0.333333
KP781	0.000000	0.011111	0.292683	0.8	0.916667	0.222222

Conditional Probability	Mathematical Representation	Probability
Given that a customer has Miles 0-49, the probability	P(KP281 Miles 0-49)	70.59%
they'll purchase KP281/KP481/KP781	P(KP481 Miles 0-49)	29.41%
	P(KP781 Miles 0-49)	0.00%
Given that a customer has Miles 50-99, the probability	P(KP281 Miles 50-99)	55.56%
they'll purchase KP281/KP481/KP781	P(KP481 Miles 50-99)	44.44%
	P(KP781 Miles 50-99)	0.00%
Given that a customer has Miles 100-149, the probability	P(KP281 Miles 100-149)	39.02%
they'll purchase KP281/KP481/KP781	P(KP481 Miles 100-149)	31.71%
	P(KP781 Miles 100-149)	29.27%
Given that a customer has Miles 150-199, the probability	P(KP281 Miles 150-199)	10.00%
they'll purchase KP281/KP481/KP781	P(KP481 Miles 150-199)	10.00%
	P(KP781 Miles 150-199)	80.00%
Given that a customer has Miles 200-249, the probability	P(KP281 Miles 200-249)	0.00%
they'll purchase KP281/KP481/KP781	P(KP481 Miles 200-249)	0.00%
	P(KP781 Miles 200-249)	100.00%

Step 5:Check the correlation among different factors


```
In [53]: # Selecting numerical columns
numerical_columns = ['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']

# Calculating correlation matrix for numerical columns only
correlation_matrix_numerical = df[numerical_columns].corr()

# Plotting heatmap for the correlation matrix of only numerical variables
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix_numerical, annot=True, cmap='coolwarm', fmt=".2f", annot_
plt.title('Correlation Heatmap (Numerical Variables Only)')
plt.show()
```



Key Observations

Correlation Level	Variables	Correlation Value	Observation
High Positive Correlations	Usage and Miles	0.77	Customers who use the treadmill more frequently tend to run/walk more miles per week.
	Fitness and Miles	0.83	Customers with higher fitness levels tend to

			run/walk more miles per week.
	Usage and Fitness	0.66	Customers who rate their fitness higher tend to use the treadmill more frequently.
	Income and Education	0.63	Higher income is moderately correlated with higher education levels.
	Income and Fitness	0.55	Higher income is moderately correlated with better self-rated fitness.
	Income and Usage	0.48	Higher income is moderately correlated with more frequent treadmill use.
Moderate Positive Correlations			
	Education and Fitness	0.44	Higher education levels are moderately correlated with higher self-rated fitness.
	Education and Usage	0.41	Higher education levels are moderately correlated with more frequent treadmill use.
Low Positive Correlations			
	Age and Income	0.51	There is a low positive correlation between age and income.
	Age and Education	0.30	Age has a low positive correlation with education levels.
	Age and Fitness	0.06	There is a very low positive correlation between age and fitness.
Low Negative Correlations			None observed in the heatmap.
Very Low to No Correlation			
	Age and Usage	0.02	Age shows almost no correlation with treadmill usage.
	Age and Miles	0.03	Age shows almost no correlation with the number of miles run/walked per week.

What does all of this means ??

Correlation Level	Variables	Correlation Value	Insights
High Positive Correlations			
	Usage and Miles	0.77	Customers who use the treadmill more frequently tend to run/walk more miles per week.
	Fitness and Miles	0.83	Customers with higher fitness levels tend to run/walk more miles per week.
	Usage and Fitness	0.66	Customers who rate their

			fitness higher tend to use the treadmill more frequently.
	Income and Education	0.63	Customers with higher income levels tend to have more education.
	Income and Fitness	0.55	Customers with higher income levels tend to rate their fitness higher.
	Income and Usage	0.48	Customers with higher income levels tend to use the treadmill more frequently.
Moderate Positive Correlations			
	Education and Fitness	0.44	Customers with more education tend to rate their fitness higher.
	Education and Usage	0.41	Customers with more education tend to use the treadmill more frequently.
Low Positive Correlations			
	Age and Income	0.51	Older customers tend to have higher income levels.
	Age and Education	0.30	Older customers tend to have more education.
	Age and Fitness	0.06	Older customers tend to rate their fitness slightly higher.
Low Negative Correlations			None observed in the heatmap.
Very Low to No Correlation			
	Age and Usage	0.02	Age shows almost no correlation with treadmill usage.
	Age and Miles	0.03	Age shows almost no correlation with the number of miles run/walked per week.

Step 6: Customer profiling and recommendation

Making customer profilings for each and every product.

Summary Statistics for Each Treadmill Product

Generate summary statistics and visualizations to profile customers who purchased each type of treadmill (KP281, KP481, KP781). This includes metrics such as age, gender, income, marital status, fitness, usage, and miles.

1. KP281 Customer Profile

Descriptive Statistics:

- **Age**
- **Gender**
- **Income**
- **Marital Status**
- **Fitness**
- **Usage**
- **Miles**

Age Distribution for KP281:

- Calculate the mean, median, and standard deviation for age.
- Plot the age distribution.

Gender Distribution for KP281:

- Count the number of males and females.
- Plot the gender distribution.

Income Distribution for KP281:

- Calculate the mean, median, and standard deviation for income.
- Plot the income distribution.

Marital Status Distribution for KP281:

- Count the number of single and partnered customers.
- Plot the marital status distribution.

Fitness Level for KP281:

- Calculate the mean, median, and standard deviation for fitness level.
- Plot the fitness level distribution.

Usage for KP281:

- Calculate the mean, median, and standard deviation for usage.
- Plot the usage distribution.

Miles for KP281:

- Calculate the mean, median, and standard deviation for miles.
- Plot the miles distribution.

```
In [54]: ##### KP281 Analysis Code:

# Filter data for KP281
kp281_data = df[df['Product'] == 'KP281']

# Age distribution statistics
age_kp281_stats = kp281_data['Age'].describe()

# Gender distribution
gender_kp281_counts = kp281_data['Gender'].value_counts()

# Income distribution statistics
income_kp281_stats = kp281_data['Income'].describe()
```

```

# Marital Status distribution
marital_status_kp281_counts = kp281_data['MaritalStatus'].value_counts()

# Fitness level distribution statistics
fitness_kp281_stats = kp281_data['Fitness'].describe()

# Usage distribution statistics
usage_kp281_stats = kp281_data['Usage'].describe()

# Miles distribution statistics
miles_kp281_stats = kp281_data['Miles'].describe()

# Plotting distributions
plt.figure(figsize=(18, 12))

plt.subplot(2, 3, 1)
plt.hist(kp281_data['Age'], bins=10, color='skyblue', edgecolor='black')
plt.title('Age Distribution for KP281')
plt.xlabel('Age')
plt.ylabel('Frequency')

plt.subplot(2, 3, 2)
gender_kp281_counts.plot(kind='bar', color=['blue', 'pink'])
plt.title('Gender Distribution for KP281')
plt.xlabel('Gender')
plt.ylabel('Count')

plt.subplot(2, 3, 3)
plt.hist(kp281_data['Income'], bins=10, color='lightgreen', edgecolor='black')
plt.title('Income Distribution for KP281')
plt.xlabel('Income')
plt.ylabel('Frequency')

plt.subplot(2, 3, 4)
marital_status_kp281_counts.plot(kind='bar', color=['orange', 'green'])
plt.title('Marital Status Distribution for KP281')
plt.xlabel('Marital Status')
plt.ylabel('Count')

plt.subplot(2, 3, 5)
plt.hist(kp281_data['Fitness'], bins=5, color='purple', edgecolor='black')
plt.title('Fitness Level Distribution for KP281')
plt.xlabel('Fitness Level')
plt.ylabel('Frequency')

plt.subplot(2, 3, 6)
plt.hist(kp281_data['Usage'], bins=10, color='red', edgecolor='black')
plt.title('Usage Distribution for KP281')
plt.xlabel('Usage (times per week)')
plt.ylabel('Frequency')

plt.tight_layout()
plt.show()

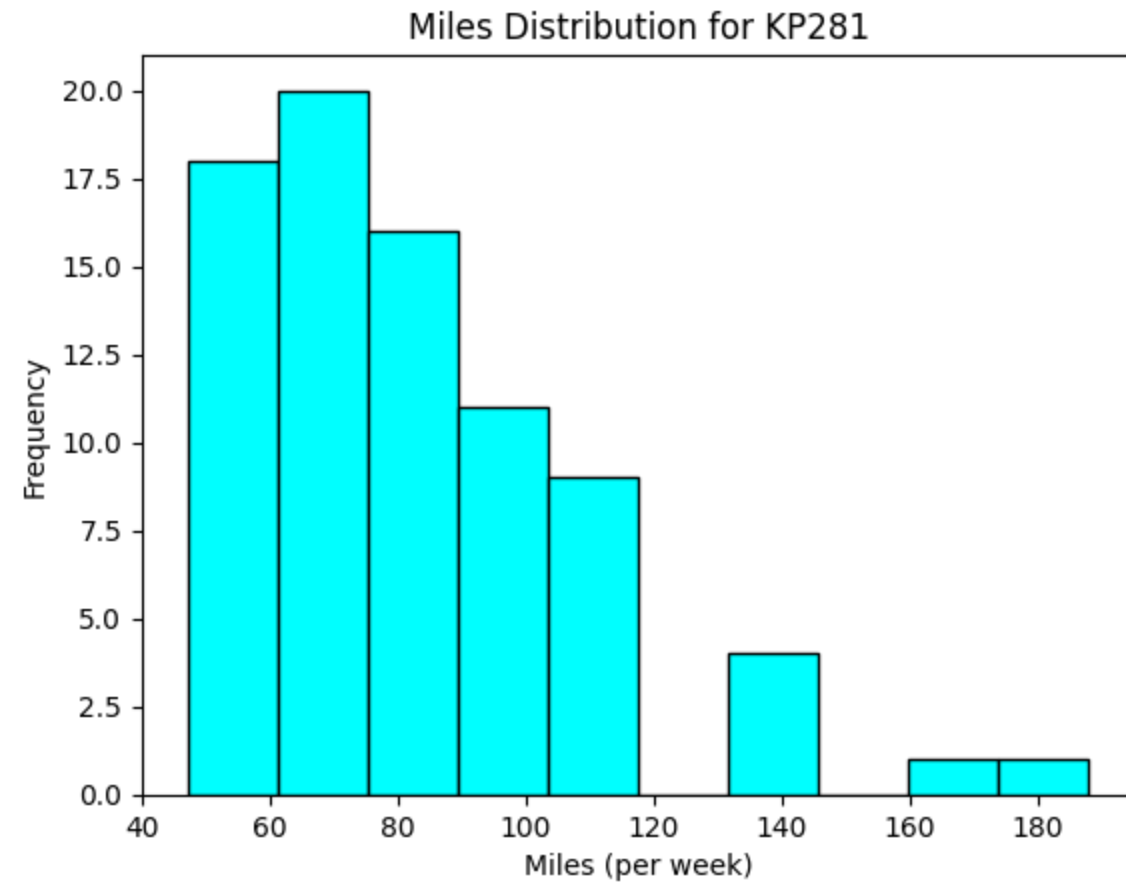
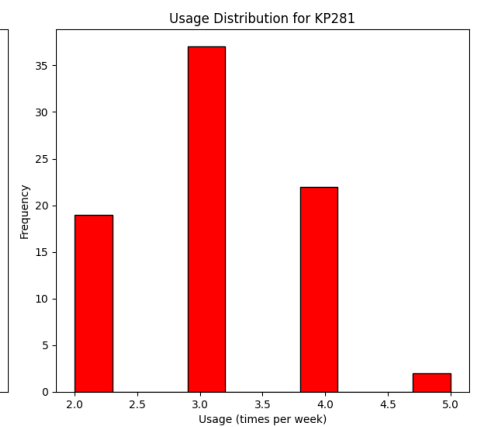
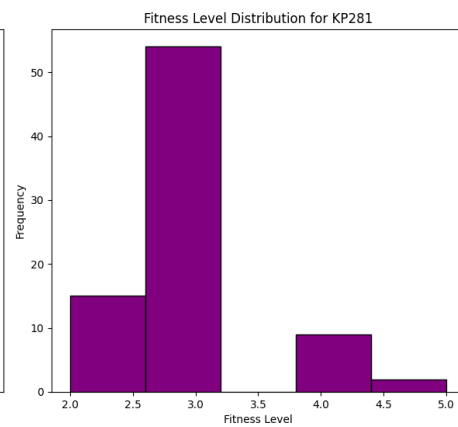
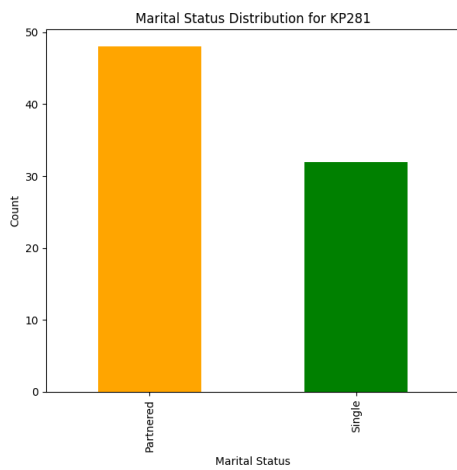
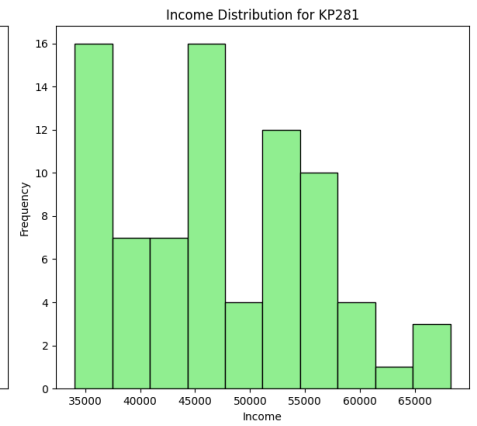
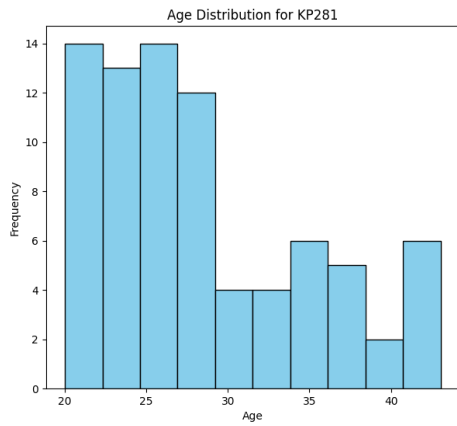
plt.figure()
plt.hist(kp281_data['Miles'], bins=10, color='cyan', edgecolor='black')
plt.title('Miles Distribution for KP281')
plt.xlabel('Miles (per week)')
plt.ylabel('Frequency')

plt.show()

print(age_kp281_stats)
print(gender_kp281_counts)
print(income_kp281_stats)

```

```
print(marital_status_kp281_counts)
print(fitness_kp281_stats)
print(usage_kp281_stats)
print(miles_kp281_stats)
```



```
count    80.000000
mean     28.427500
std      6.678313
```

```

min      20.000000
25%      23.000000
50%      26.000000
75%      33.000000
max      43.050000
Name: Age, dtype: float64
Gender
Male      40
Female    40
Name: count, dtype: int64
count      80.000000
mean     46584.311250
std      8813.246103
min     34053.150000
25%     38658.000000
50%     46617.000000
75%     53439.000000
max     68220.000000
Name: Income, dtype: float64
MaritalStatus
Partnered  48
Single     32
Name: count, dtype: int64
count      80.000000
mean        2.975000
std         0.635948
min         2.000000
25%         3.000000
50%         3.000000
75%         3.000000
max         5.000000
Name: Fitness, dtype: float64
count      80.000000
mean        3.087500
std         0.782624
min         2.000000
25%         3.000000
50%         3.000000
75%         4.000000
max         5.000000
Name: Usage, dtype: float64
count      80.000000
mean       83.125000
std       28.391198
min       47.000000
25%       66.000000
50%       85.000000
75%       94.000000
max      188.000000
Name: Miles, dtype: float64

```

```

In [55]: # KP481 Analysis Code:

# Filter data for KP481
kp481_data = df[df['Product'] == 'KP481']

# Age distribution statistics
age_kp481_stats = kp481_data['Age'].describe()

# Gender distribution
gender_kp481_counts = kp481_data['Gender'].value_counts()

# Income distribution statistics

```

```

income_kp481_stats = kp481_data['Income'].describe()

# Marital Status distribution
marital_status_kp481_counts = kp481_data['MaritalStatus'].value_counts()

# Fitness level distribution statistics
fitness_kp481_stats = kp481_data['Fitness'].describe()

# Usage distribution statistics
usage_kp481_stats = kp481_data['Usage'].describe()

# Miles distribution statistics
miles_kp481_stats = kp481_data['Miles'].describe()

# Plotting distributions
plt.figure(figsize=(18, 12))

plt.subplot(2, 3, 1)
plt.hist(kp481_data['Age'], bins=10, color='skyblue', edgecolor='black')
plt.title('Age Distribution for KP481')
plt.xlabel('Age')
plt.ylabel('Frequency')

plt.subplot(2, 3, 2)
gender_kp481_counts.plot(kind='bar', color=['blue', 'pink'])
plt.title('Gender Distribution for KP481')
plt.xlabel('Gender')
plt.ylabel('Count')

plt.subplot(2, 3, 3)
plt.hist(kp481_data['Income'], bins=10, color='lightgreen', edgecolor='black')
plt.title('Income Distribution for KP481')
plt.xlabel('Income')
plt.ylabel('Frequency')

plt.subplot(2, 3, 4)
marital_status_kp481_counts.plot(kind='bar', color=['orange', 'green'])
plt.title('Marital Status Distribution for KP481')
plt.xlabel('Marital Status')
plt.ylabel('Count')

plt.subplot(2, 3, 5)
plt.hist(kp481_data['Fitness'], bins=5, color='purple', edgecolor='black')
plt.title('Fitness Level Distribution for KP481')
plt.xlabel('Fitness Level')
plt.ylabel('Frequency')

plt.subplot(2, 3, 6)
plt.hist(kp481_data['Usage'], bins=10, color='red', edgecolor='black')
plt.title('Usage Distribution for KP481')
plt.xlabel('Usage (times per week)')
plt.ylabel('Frequency')

plt.tight_layout()
plt.show()

plt.figure()
plt.hist(kp481_data['Miles'], bins=10, color='cyan', edgecolor='black')
plt.title('Miles Distribution for KP481')
plt.xlabel('Miles (per week)')
plt.ylabel('Frequency')

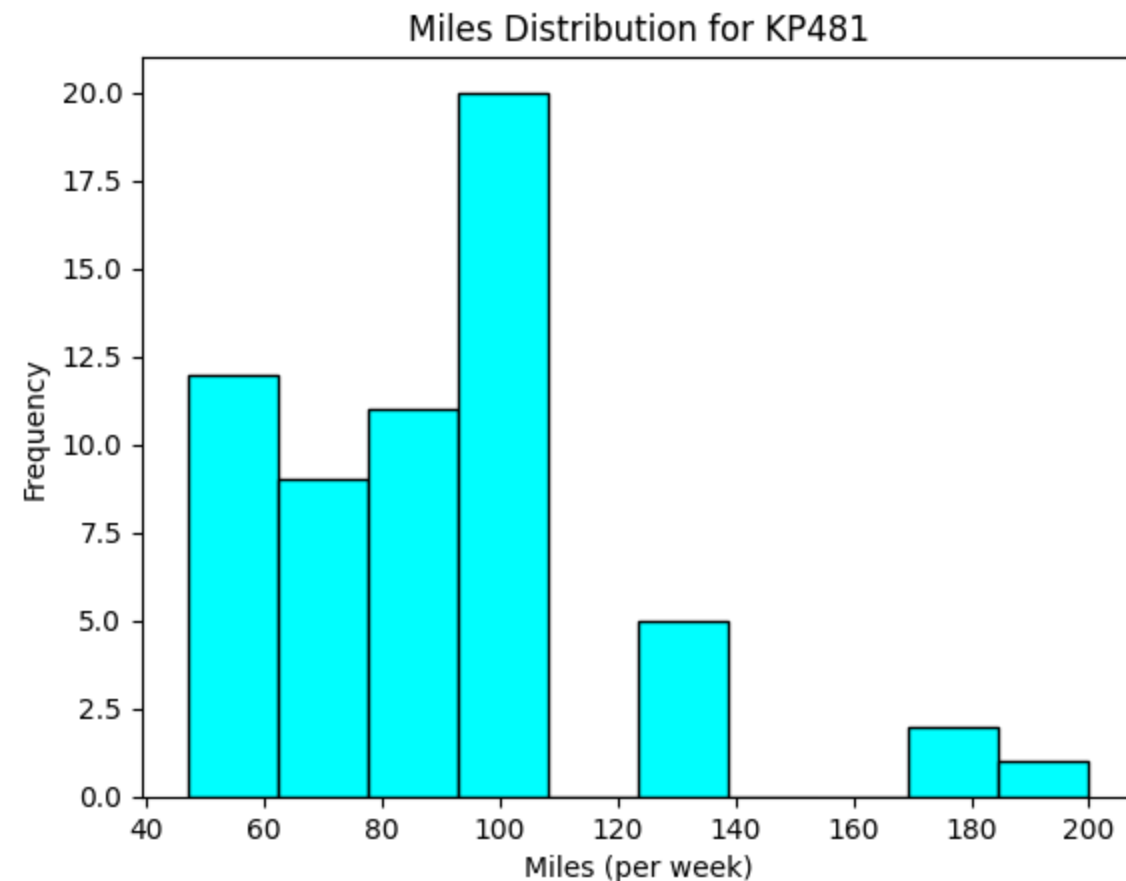
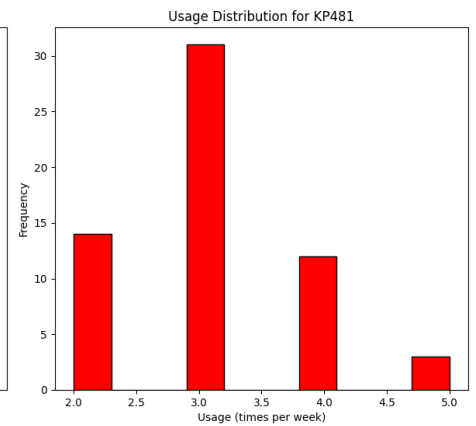
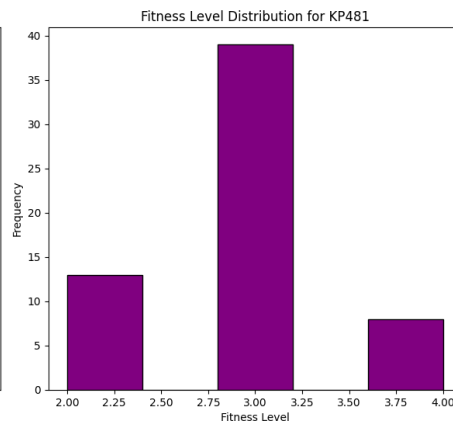
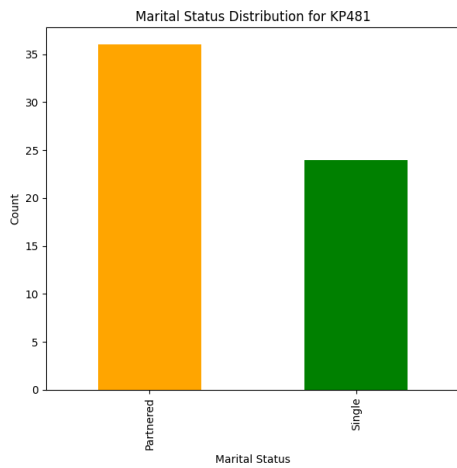
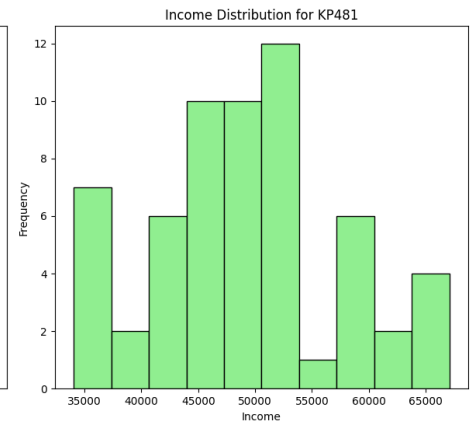
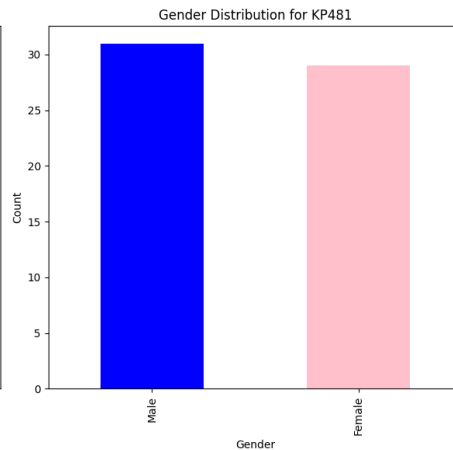
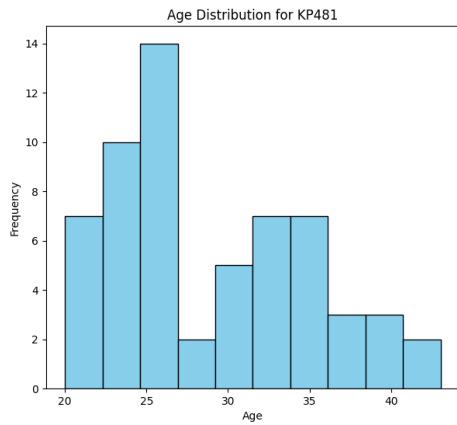
plt.show()

print(age_kp481_stats)
print(gender_kp481_counts)

```



```
print(income_kp481_stats)
print(marital_status_kp481_counts)
print(fitness_kp481_stats)
print(usage_kp481_stats)
print(miles_kp481_stats)
```



```
count    60.000000
mean     28.801667
```

```

std      6.327830
min      20.000000
25%      24.000000
50%      26.000000
75%      33.250000
max      43.050000
Name: Age, dtype: float64
Gender
Male      31
Female    29
Name: count, dtype: int64
count      60.000000
mean     49046.607500
std      8517.583361
min     34053.150000
25%     44911.500000
50%     49459.500000
75%     53439.000000
max     67083.000000
Name: Income, dtype: float64
MaritalStatus
Partnered    36
Single       24
Name: count, dtype: int64
count      60.000000
mean        2.916667
std         0.590652
min         2.000000
25%         3.000000
50%         3.000000
75%         3.000000
max         4.000000
Name: Fitness, dtype: float64
count      60.000000
mean        3.066667
std         0.799717
min         2.000000
25%         3.000000
50%         3.000000
75%         3.250000
max         5.000000
Name: Usage, dtype: float64
count      60.000000
mean       88.500000
std       31.322543
min       47.000000
25%       64.000000
50%       85.000000
75%      106.000000
max      200.000000
Name: Miles, dtype: float64

```

In [56]: *# 3. KP781 Customer Profile*

```

# Filter data for KP781
kp781_data = df[df['Product'] == 'KP781']

# Age distribution statistics
age_kp781_stats = kp781_data['Age'].describe()

# Gender distribution
gender_kp781_counts = kp781_data['Gender'].value_counts()

```

```

# Income distribution statistics
income_kp781_stats = kp781_data['Income'].describe()

# Marital Status distribution
marital_status_kp781_counts = kp781_data['MaritalStatus'].value_counts()

# Fitness level distribution statistics
fitness_kp781_stats = kp781_data['Fitness'].describe()

# Usage distribution statistics
usage_kp781_stats = kp781_data['Usage'].describe()

# Miles distribution statistics
miles_kp781_stats = kp781_data['Miles'].describe()

# Plotting distributions
plt.figure(figsize=(18, 12))

plt.subplot(2, 3, 1)
plt.hist(kp781_data['Age'], bins=10, color='skyblue', edgecolor='black')
plt.title('Age Distribution for KP781')
plt.xlabel('Age')
plt.ylabel('Frequency')

plt.subplot(2, 3, 2)
gender_kp781_counts.plot(kind='bar', color=['blue', 'pink'])
plt.title('Gender Distribution for KP781')
plt.xlabel('Gender')
plt.ylabel('Count')

plt.subplot(2, 3, 3)
plt.hist(kp781_data['Income'], bins=10, color='lightgreen', edgecolor='black')
plt.title('Income Distribution for KP781')
plt.xlabel('Income')
plt.ylabel('Frequency')

plt.subplot(2, 3, 4)
marital_status_kp781_counts.plot(kind='bar', color=['orange', 'green'])
plt.title('Marital Status Distribution for KP781')
plt.xlabel('Marital Status')
plt.ylabel('Count')

plt.subplot(2, 3, 5)
plt.hist(kp781_data['Fitness'], bins=5, color='purple', edgecolor='black')
plt.title('Fitness Level Distribution for KP781')
plt.xlabel('Fitness Level')
plt.ylabel('Frequency')

plt.subplot(2, 3, 6)
plt.hist(kp781_data['Usage'], bins=10, color='red', edgecolor='black')
plt.title('Usage Distribution for KP781')
plt.xlabel('Usage (times per week)')
plt.ylabel('Frequency')

plt.tight_layout()
plt.show()

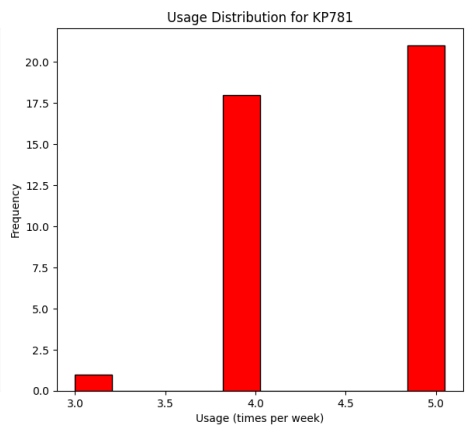
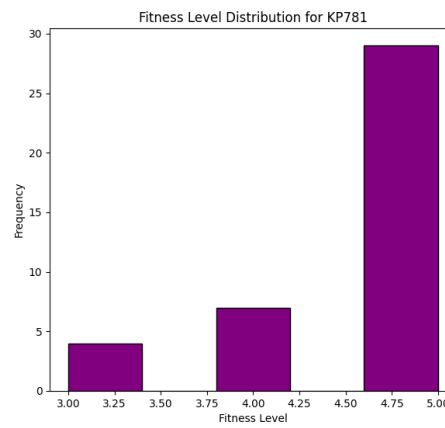
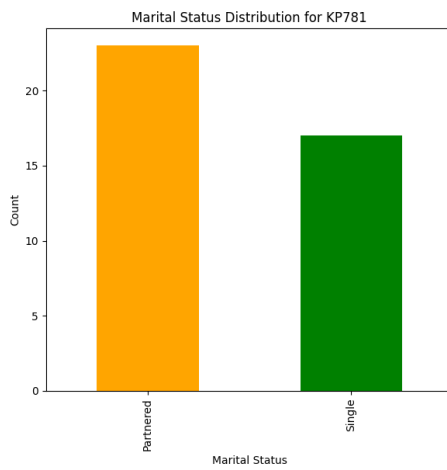
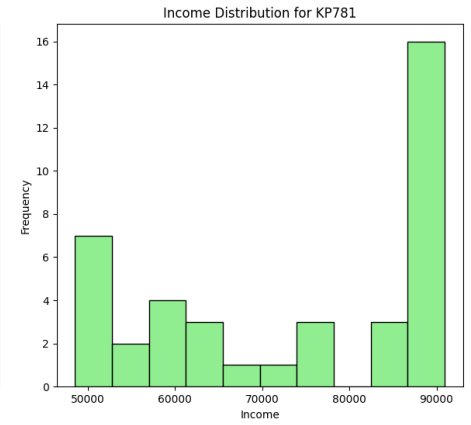
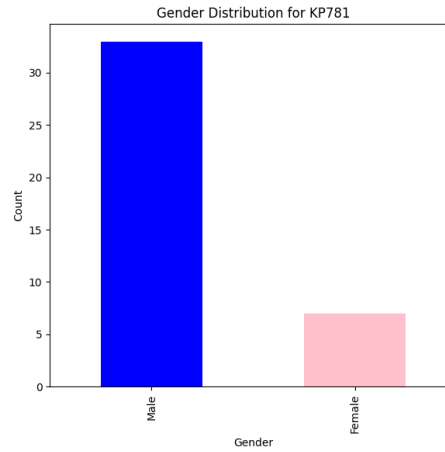
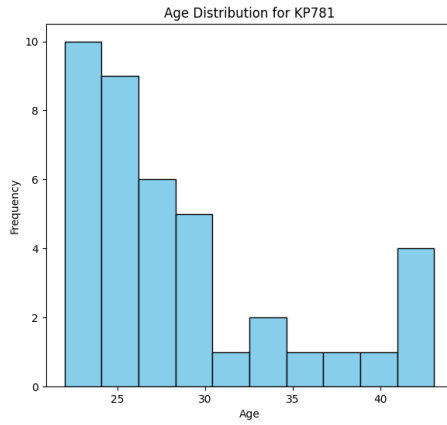
plt.figure()
plt.hist(kp781_data['Miles'], bins=10, color='cyan', edgecolor='black')
plt.title('Miles Distribution for KP781')
plt.xlabel('Miles (per week)')
plt.ylabel('Frequency')

plt.show()

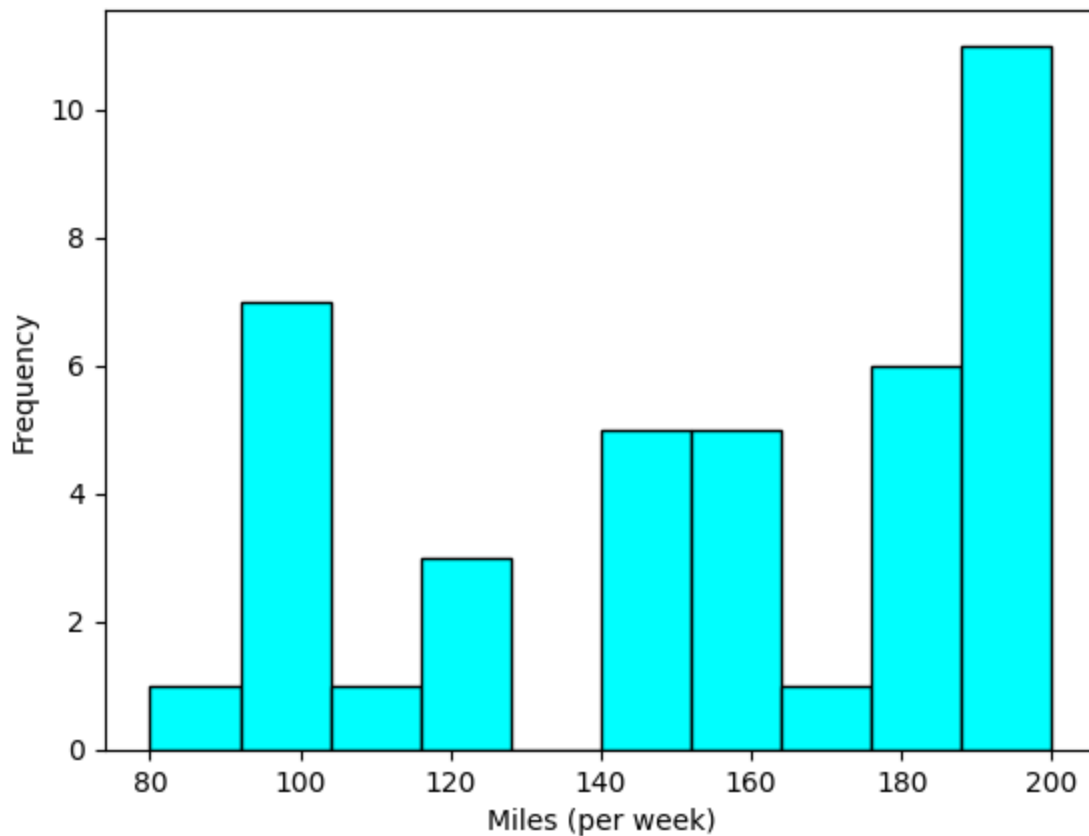
print(age_kp781_stats)

```

```
print(gender_kp781_counts)
print(income_kp781_stats)
print(marital_status_kp781_counts)
print(fitness_kp781_stats)
print(usage_kp781_stats)
print(miles_kp781_stats)
```



Miles Distribution for KP781



count 40.000000

```
mean      28.828750
std       6.296182
min       22.000000
25%      24.750000
50%      27.000000
75%      30.250000
max       43.050000
Name: Age, dtype: float64
Gender
Male      33
Female    7
Name: count, dtype: int64
count     40.000000
mean     73908.281250
std     16572.164368
min     48556.000000
25%     58204.750000
50%     76568.500000
75%     90886.000000
max     90948.250000
Name: Income, dtype: float64
MaritalStatus
Partnered 23
Single    17
Name: count, dtype: int64
count     40.000000
mean       4.625000
std        0.667467
min        3.000000
25%        4.000000
50%        5.000000
75%        5.000000
max        5.000000
Name: Fitness, dtype: float64
count     40.000000
mean       4.511250
std        0.565401
min        3.000000
25%        4.000000
50%        5.000000
75%        5.000000
max        5.050000
Name: Usage, dtype: float64
count     40.000000
mean     155.900000
std       39.073763
min       80.000000
25%      120.000000
50%      160.000000
75%      200.000000
max       200.000000
Name: Miles, dtype: float64
```

KP281 Customer Profile

Feature	Insight
Affordability and Popularity	KP281 is the most popular and affordable entry-level treadmill, making it a top choice for many customers.
Gender	Equally popular among both male and female customers.
Distance Covered	Customers typically cover 70 to 90 miles on this treadmill.
Usage Frequency	Used 3 to 4 times a week, indicating a moderate usage pattern.

Fitness Rating	Most customers rate their fitness as "Average-Rating 3," suggesting it's favored by those starting their fitness journey.
Age Group	Attracts a wide range of ages from younger(20s to 30s) to elder beginners.
Marital Status	Single females and partnered males are the predominant buyers, more than single males.
Income Range	Preferred by customers with an income between \$39,000 and \$53,000.

KP481 Customer Profile

Feature	Insight
Affordability and Popularity	KP481 is a mid-range product, popular among regular exercisers looking for more features than the entry-level model.
Gender	Equally popular among both male and female customers. (slightly more in males)
Distance Covered	Customers typically cover around 50-110 miles per week.
Usage Frequency	Used 3 to 4 times a week, indicating consistent usage.
Fitness Rating	Customers rate their fitness between 3 and 4, showing it's used by those with moderate to high fitness levels.
Age Group	Attracts middle-aged adults (25,30-35) years.
Marital Status	Mostly purchased by partnered individuals, suggesting a family-oriented demographic.
Income Range	Preferred by those with an income between \$45,000 and \$70,000.

KP781 Customer Profile

Feature	Insight
Affordability and Popularity	KP781 is a high-end treadmill with advanced features, appealing to serious runners and fitness enthusiasts.
Gender	KP781 is more popular among male customers.
Distance Covered	Customers typically cover more than 90 miles, more around 100-120 miles per week indicating high usage for serious fitness goals.
Usage Frequency	Used more frequently, 4 to 6 times a week, by highly committed individuals.
Fitness Rating	Customers rate their fitness higher, often "Good" to "Excellent," indicating it attracts those in better physical shape.
Age Group	Attracts customers primarily in their mid-20s to early 30s and above 40s
Marital Status	More preferred by Partenered , however almost equally popular among single and partnered individuals, indicating it's chosen by a wide demographic.
Income Range	Preferred by customers with higher incomes, typically between \$70,000 and \$90,000, indicating it appeals to more affluent customers.

Recommendations:

1. **Encourage Female Engagement:** Most women plan on using the treadmill less frequently than men. Create marketing campaigns that highlight features appealing to women, such as easy-to-use programs and safety features.

2. **Promote KP281 as an Affordable Option:**

- **Target Audience:** Younger individuals, beginners, or those with lower to mid-income levels (40,000–80,000).
- **Marketing Strategy:** Highlight affordability, accessibility, and beginner-friendly features.

3. **Position KP481 for Middle-Income Regular Users:**

- **Target Audience:** Middle-income customers with moderate fitness levels (50,000–70,000).
- **Marketing Strategy:** Emphasize value for money, durability, and suitability for regular use.

4. **Market KP781 to Affluent, Fitness-Conscious Individuals:**

- **Target Audience:** Higher-income individuals (60,000–90,000), serious runners, and those who exercise intensively.
- **Marketing Strategy:** Focus on advanced features, high performance, and durability. Use athlete endorsements and influencer marketing.

5. **Age-Specific Campaigns:**

- **For Younger Customers:** Design products and marketing strategies that cater to younger individuals, particularly those around 25 years old.
- **For Older Customers:** Research and develop products that meet the needs of individuals over 50, considering health benefits and usability.

6. **Income-Based Financing Options:**

- **KP281 & KP481:** Offer promotions and financing options to target middle-income customers.
- **KP781:** Highlight the advanced features and benefits to wealthier individuals.

7. **Fitness Level Programs:**

- **For Beginners:** Develop fitness programs and support to enhance appeal for KP281 and KP481.
- **For Advanced Users:** Emphasize advanced training programs and performance tracking features for KP781.

8. **Usage-Based Product Positioning:**

- **Moderate Usage (KP281 & KP481):** Market these models as reliable and user-friendly for moderate exercise routines.
- **High Usage (KP781):** Target serious runners or frequent users, emphasizing the treadmill's durability and high performance.

9. **Tailored Marketing by Marital Status:**

- **Partnered Individuals:** Focus on premium products like KP781, which show a slight preference among partnered customers.
- **Single Individuals:** Tailor marketing strategies to highlight affordability and value for products like KP281 and KP481.

10. **Segment-Based Offers:**

- **Fitness Level Segmentation:** Offer special deals for customers looking to improve from fitness levels 2 to 3 or 3 to 4.

- **Usage Frequency Segmentation:** Create different programs or features for casual users (1-3 times/week) and frequent users (4-6 times/week).
-

Additional Recommendations:

1. Influencer Marketing

- **Leverage Fitness Influencers:** Collaborate with popular fitness influencers and athletes who resonate with your target demographic. These influencers can demonstrate the benefits and features of AeroFit treadmills in their workouts.
- **International Athletes:** Use international athletes as influencers to enhance global appeal and credibility. These athletes can showcase how AeroFit treadmills support their intensive training regimens promoting KP781.
- **Testimonials and Reviews:** Use authentic reviews and testimonials from influencers to build credibility and trust with potential customers.

2. TikTok and Instagram Reels

- **Engaging Content:** Create short, engaging videos showcasing different workout routines, treadmill features, and success stories. Highlight unique features of each treadmill model.
- **Challenges and Hashtags:** Launch fitness challenges with branded hashtags to encourage user-generated content. For example, a "30-day treadmill challenge" can motivate users to share their progress.

3. Female-Tailored Marketing

- **Highlight Specific Features:** Emphasize features that cater to female users, such as easy-to-use interfaces, safety features, and programs tailored for different fitness levels.
- **Women-Focused Campaigns:** Run campaigns during Women's Health Month or other relevant occasions, encouraging women to prioritize fitness. Use relatable female fitness influencers to endorse products.
- **Community Building:** Create online communities or forums where women can share their fitness journeys and tips, fostering a supportive environment.

4. Discount Offers

- **Seasonal Promotions:** Offer discounts during major sales events like Black Friday, New Year, or back-to-school periods.
- **Referral Discounts:** Encourage existing customers to refer friends and family by offering discounts on future purchases for both the referrer and the referee.
- **Bundle Deals:** Create bundles that include accessories like workout mats, water bottles, or heart rate monitors with treadmill purchases.

5. Upsell

- **Upgrade Programs:** Offer existing customers trade-in deals or discounts on newer, higher-end models like the KP781. Highlight the advanced features and additional benefits.
- **Membership Packages:** Introduce membership plans that include regular maintenance, exclusive workout content, and priority customer service for an additional fee.
- **Extended Warranties and Services:** Upsell extended warranties, personalized training plans, and on-demand virtual workouts.

6. Upsell Young People for Partnered, More Educated

- **Targeted Ads:** Use data to target ads specifically to young, partnered individuals with higher education levels, highlighting features that appeal to their lifestyle and fitness goals.
- **Social Proof:** Use testimonials and endorsements from young, educated couples who use AeroFit products together to maintain their fitness.
- **Special Offers for Couples:** Provide discounts for couples who purchase two treadmills or fitness products together, promoting a shared fitness journey.
- **Showcase Products in Expos:** Participate in fitness expos and health fairs to showcase AeroFit products. Create engaging booths that attract 25 to 30-year-olds, especially those who are dating and want to look good. Highlight how regular exercise with AeroFit treadmills can help them achieve their fitness goals and enhance their physical appearance.
- **Run Ads on Tinder:** Place ads on Tinder and similar dating platforms where young, single, active individuals are likely to be found. Highlight the benefits of AeroFit treadmills for achieving fitness goals and looking good, which appeals to those looking to improve their physical appearance for dating.

7. Daily Newsletter

- **Fitness Tips and Tricks:** Include daily fitness tips, workout routines, and health advice in the newsletter to keep your audience engaged and motivated.
- **Product Spotlights:** Feature different treadmill models, highlighting their unique features and benefits.
- **Success Stories:** Share customer success stories and testimonials to inspire readers and build a community.
- **Exclusive Offers:** Provide newsletter subscribers with exclusive discount codes, early access to sales, and updates on new product launches.
- **Interactive Content:** Incorporate interactive content like polls, quizzes, and Q&A sessions with fitness experts to keep subscribers engaged.

Implementing these strategies will enhance AeroFit's marketing efforts, making them more appealing and effective across different customer segments.

-----The End-----